



# Cybersecurity Guidelines for **[Space Systems]**

Sept. 2025





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## History of Revision

[illegible]

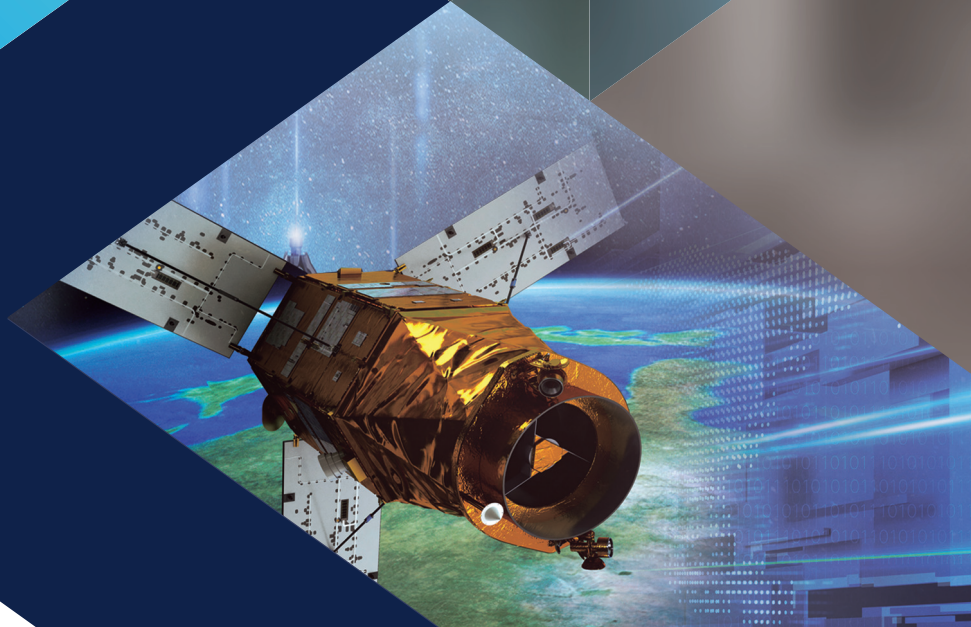


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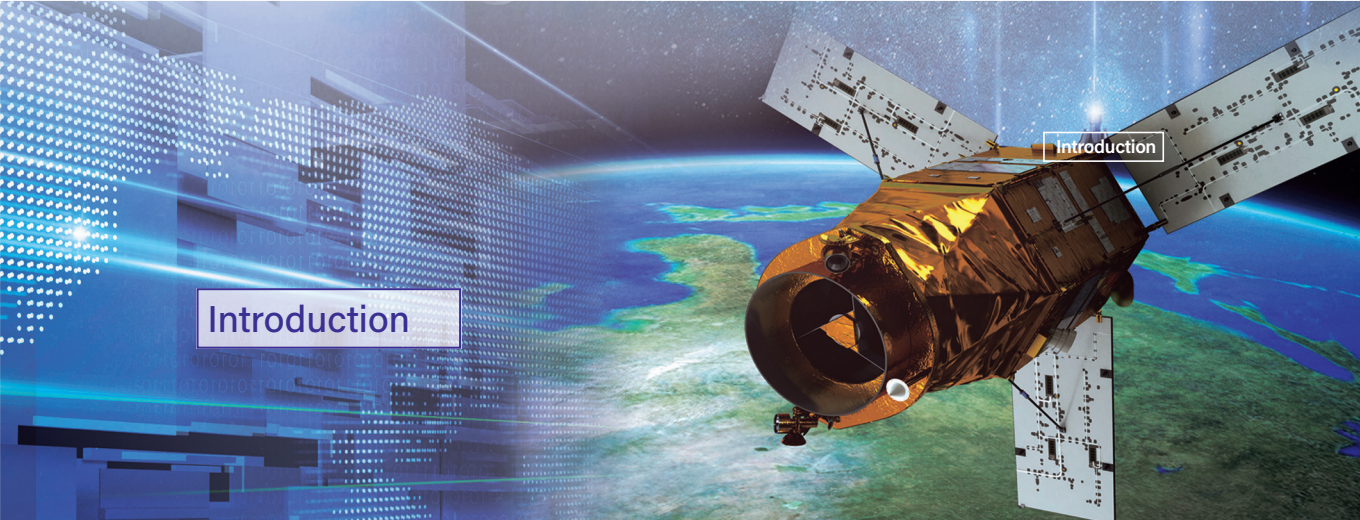
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# Introduction







## Introduction

This document presents cybersecurity considerations and countermeasures to be applied throughout the entire life cycle of space systems – from design and development, then to operation and utilization, and finally to decommissioning.

Chapter 1 defines terminology and standard models for space systems, and Chapter 2 presents cybersecurity threats to space systems and common attack scenarios.

Chapter 3 describes the security measures for each component of a space system, such as the ground station, satellite network, and satellite, across the different stages of the space system's life cycle.

The appendix includes security checklists for space system components to be completed in different stages of life cycle.

Space systems can utilize various security countermeasures based on their mission functions and the services they provide. As a result, organizations that design and operate space systems have encountered challenges in implementing effective cybersecurity measures.

Hence, this document is intended to offer practical guidance for organizations engaged in the design, development, operation, and utilization of space systems, specifically to assist in the preparation and implementation of cybersecurity measures.

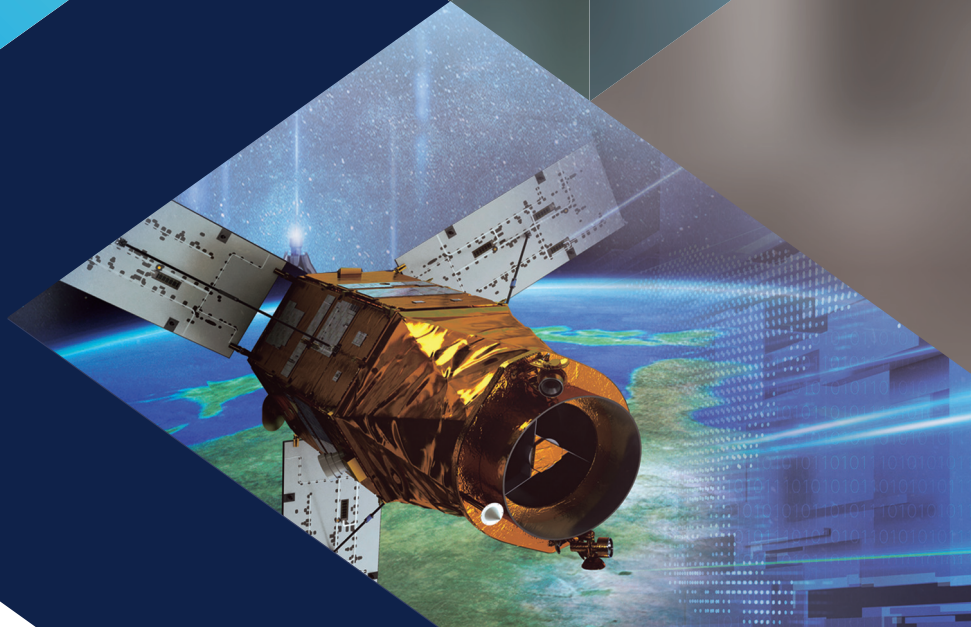
They will be continuously updated to reflect relevant developments in a timely manner.





## Chapter 1

# Overview







## Section 1

## Background

In recent years, space has rapidly emerged as a critical area of national economy and security, as satellites and other space systems are increasingly utilized in essential sectors such as defense, communications, and transportation.

Countries worldwide are increasingly prioritizing space technology, driven by the impactful activities of SpaceX's Starlink in the Russo-Ukrainian War. These effects highlight the importance of space systems.

So far, individuals involved in space systems have relied on security by obscurity, ensuring security by not revealing these systems' design, implementation, and operation to the outside world.

This secrecy may yield temporary benefits, but this approach gradually becomes less secure as hackers intensify their interest and sensitive information gets more susceptible to exposure.

With the advent of the New Space Age, the space industry is expanding and the proliferation of services such as low-orbit satellite communications is bringing space technology closer to the country and its people alike. This naturally increases hackers' interest, making space systems a key target for cyberattacks. There is now a greater range of space systems for cybercriminals to target using

increasingly sophisticated techniques. As national security can be impacted if a country's strategic assets in military, navigation, communications, science, earth observation, and space exploration become compromised by cyberattacks, change in space cybersecurity strategy is required now.

This document provides a description of cyber threats to space systems, including ground stations, satellite networks, and satellites, to enlist vigilance, and presents security measures that can allow entities with New Space Age aspirations to safely develop and operate space systems.

## Section 2 Terminology

As entities involved with space systems can observe varying definitions of terms used, this section defines the terms used as follows.

**Cyberattacks, cyber threats** Activities or associated threats in the invasion, disruption, paralysis, or destruction of information and communications systems or networks by electronic means such as hacking, computer viruses, ransomware, DDoS<sup>1)</sup>, and electromagnetic waves, or in the forgery, alteration, damage, or stealing of information

**Space system** A system consisting of several interacting components for the execution of missions in space environments, such as space explorations and satellite operations; examples include launch vehicles, ground stations, satellite network, satellite buses, payloads, communication equipment, satellite developers and manufacturers.

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1) DDoS : Distributed Denial of Service

**Artificial space object** An object designed and made for use in space; examples include launch vehicles, satellites, spacecraft, and their components

**Launch vehicle** An artificial space object which drives a satellite or spacecraft into space using a self-propelled engine

**Space accident** Malfunctions, crashes, collisions, and explosions resulting from cyberattacks during the launch (launch preparations, test launches, and failed launches included) or operation of artificial space objects

**Satellite data** Videos, voices, sounds, and other data, including combinations of such data (processing or utilization of such data combinations included), obtained using a satellite

**Space risk** A risk involving the crash, collision, or cyberattacks on space objects in outer space

**Satellite database** A collection of systematically organized and processed satellite data to be searched and utilized by data users

**Spacecraft** A launch vehicle, space shuttle, satellite, or manned or unmanned spacecraft capable of flying in or outside Earth's atmosphere, or other devices that can be used for space flight

**Space object launch** The launch of a space object by an entity under national authority, including launch preparations, test launches, and failed launches

**Electromagnetic interference** The generation of electromagnetic waves, the emission of electromagnetic waves from equipment and parts (spread of electromagnetic energy in a space), or the occurrence of conduction (flow of electromagnetic energy through a power line) impairing the performance of other equipment and parts

**Satellite asset** An artificial space object or an associated facility or system

**Security-related space data** Any of the below data related to national security

a. Data on satellite assets

- b. Data on space accidents, satellite data, and space risks
- c. Data on threats to the security of satellite assets posed by North Korea, foreign countries and nationals, foreign groups, non-state actors, or Koreans linked to such entities

**Information security** Any of the activities of an administrative, physical, or technical nature performed to prevent the leak, forgery, alteration, or damage of data collected, processed, stored, searched, or transmitted through information and communications systems and networks for the functioning of various institutions

**Artificial satellite** A man-made device launched into space for a specific purpose, typically orbiting a celestial body

**Satellite** A concept which includes both natural satellites and artificial satellites; although the term “satellite” can refer to any object orbiting a celestial body, in this document it refers to artificial satellites, unless otherwise indicated

An artificial satellite, including a satellite bus and payload

**Satellite bus** A system which constitutes the basic structure and functional elements of a satellite as needed to execute missions

**Payload** Equipment attached to a satellite allowing the satellite to execute missions; examples include equipment for communications, imaging, or climate observation

**Ground station** A facility or institution which performs control and management from Earth to enable a space system to function in space; categorized as a satellite control center, satellite data center, or satellite communication center

**Satellite network** The communication network on which a satellite and ground station operate, including satellite frequencies and orbits

**Satellite frequency** A frequency used on a satellite network

**Satellite orbit** The position or trajectory of a satellite

**Satellite operation system** A ground station system for the operation of a satellite

**Satellite operation center** A facility or institution holding a satellite operation system

As facilities or institutions operating satellites typically also engage in control activities, the term “satellite control center” is also used

**Satellite data center** A facility or institution utilizing mission data sent from the payload of a satellite; functions include the transmission and utilization of mission data and satellite data

Using an independently owned ground station system, satellite data centers receive mission data directly from a satellite payload or through a satellite operation center on a wired network

**Satellite communication center** A facility responsible for the operation, management, and maintenance of satellite communication systems; provides satellite communication services

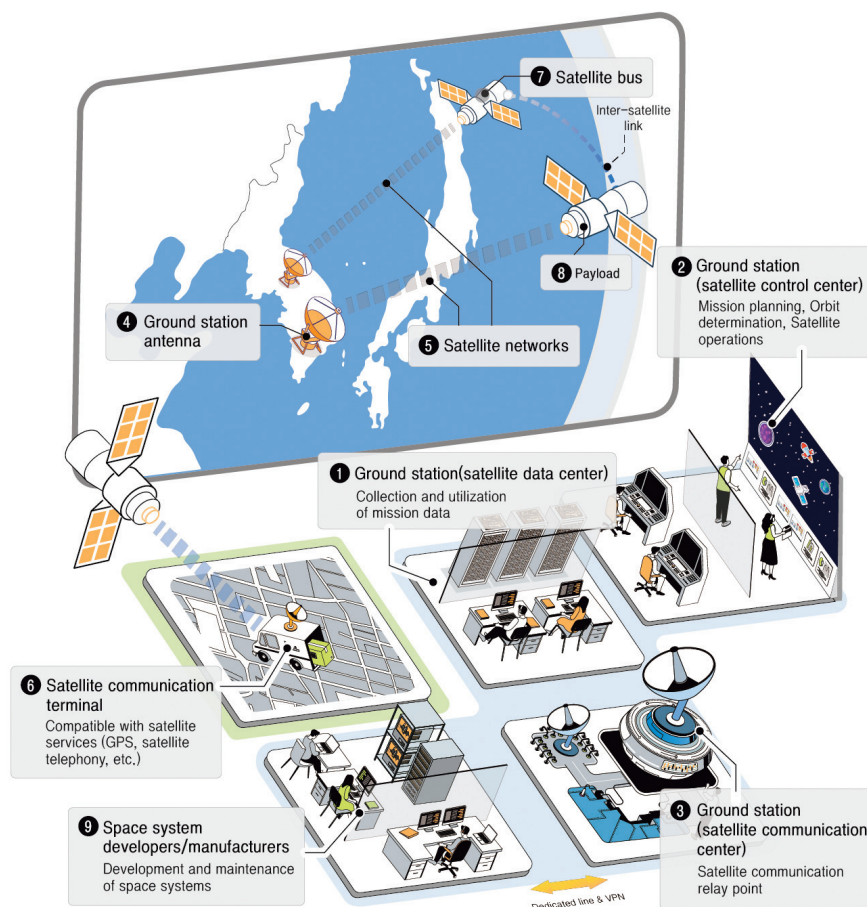
## Section 3

## Standard Model for Space System

To support understanding of the security threats and measures concerning space systems described in this document, the standard model for space system shown in [Fig. 1-1] serves as a generalization of the space systems of various configurations in existence.



Fig. 1-1 | Standard Model for Space System



The three major categories of space system are “ground station”, “satellite network”, and “satellite”. Based on their functions, ground stations are divided into satellite data center(①), satellite control center(②), and satellite communication center(③). Satellite data centers equipped with satellite communication equipment such as a ground station antenna(④) receive mission data(e.g. images, observation data, etc.) directly from a satellite payload. Satellite communication centers without satellite communication equipment receive mission data through

a satellite control center on a wired network. Satellite communication centers are responsible for transmitting and receiving satellite signals and switching between wireless and wired connections. Satellite control centers perform control and operational activities that include the planning of missions and the transmission of control commands to satellite bodies.

Satellite networks(❺) enable inter-satellite link between satellites or ground-satellite communication, as well as wired communication among ground stations, or among ground stations and space system developers and manufacturers. Users use a satellite communication terminal(❻) to access satellite communication services such as GPS and satellite telephony.

A satellite comprises a satellite bus(❼) and payload(❽). The range of purposes satellites serve includes military, navigation, communications, science, earth observation, and space exploration. The specifics are as follows.

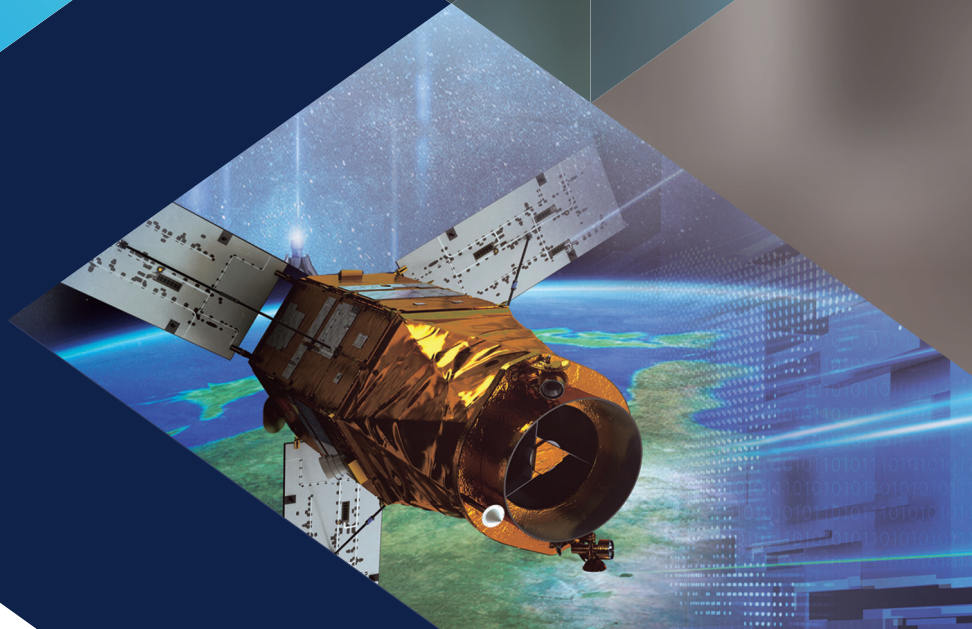
- Military satellite: Reconnaissance, surveillance, military communications
- Navigation satellite: Provision of location and time data such as that provided through GPS, GLONASS, and Galileo
- Communication satellite: Transmission of voices, images, and other data
- Scientific satellite: Research in astronomy, physics, and earth science
- Earth observation satellite: Image Acquisition, Monitoring of weather, climate, environmental changes, resources, etc.
- Space exploration satellite: Lunar orbiters, Mars rovers, Space Environment Research, etc.

Developers and manufacturers involved in the development and maintenance of space systems (❾) are also included in the above standard model as they are also exposed to cyber threats and can affect an entire space system.



## Chapter 2

# Cyber threats to Space Systems







## Chapter 2

# Cyber threats to Space Systems

This chapter discusses significant global cybersecurity threats targeting space systems. Based on these threats, it identifies potential risks stemming from cyberattack vulnerabilities and outlines likely cybersecurity threat scenarios.

## Section 1

### Examples of Threats to Space Systems

Cyberattacks on space systems such as ground stations and satellites have occurred in different parts of the world over the last decade. As interest in space has increased recently, so has the number of victims.

Before the start of the Russo-Ukrainian War in February 2022, American communications company Viasat was subjected to a cyberattack taking advantage of vulnerabilities in Viasat's satellite communication modems, resulting in a partial communication network failure affecting Ukraine and other parts of Europe. The incident was a watershed resulting in global recognition of the importance of space in national security.

In June 2023, an internet server of the JAXA<sup>2)</sup> was hacked and personal information of personnel and confidential data entrusted by third-party partners were stolen from the JAXA's internal networks.

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2) JAXA : Japan Aerospace Exploration Agency

Cyberattacks on space systems have also occurred in Korea. In 2023 a national satellite network operated by the Korean government was hacked, and a national satellite control center received a cyberattack. The purpose of such cyberattacks is to steal technology or confidential information or cause disruptions in society.

[Table 1] Past Major Threats to Space Systems

| Date      | Cyber threat                                                                                                                                                                                                                                                                                                     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nov. 2014 | <ul style="list-style-type: none"> <li>• Hacking attack on the US National Oceanic and Atmospheric Administration's (NOAA) satellite</li> <li>- Temporary disruption of access to the NOAA's climate observation network</li> </ul>                                                                              |
| Sep. 2015 | <ul style="list-style-type: none"> <li>• Use of a commercial satellite in a series of hacking attacks</li> <li>- Hacking of an unencrypted satellite network, hiding a hacking attack, and etc.</li> </ul>                                                                                                       |
| Oct. 2018 | <ul style="list-style-type: none"> <li>• Hacking attack on a NASA server</li> <li>- Breach of NASA personnel data</li> </ul>                                                                                                                                                                                     |
| Feb. 2022 | <ul style="list-style-type: none"> <li>• Hacking attack on US satellite communication company Viasat</li> <li>- Paralysis of Viasat's satellite modem in the hours leading up to the start of the Russo-Ukrainian War resulted in a partial communication failure affecting Europe</li> </ul>                    |
| May 2022  | <ul style="list-style-type: none"> <li>• Hacking attempt on Starlink</li> <li>- Radio jamming and hacking attempts on Starlink</li> </ul>                                                                                                                                                                        |
| Jun. 2023 | <ul style="list-style-type: none"> <li>• Partial disruption of a Russian military satellite network</li> <li>- Temporary disruption by malware attack exploiting software vulnerabilities on Russian satellite telecommunications provider Dozor-Teleport, which operates Russian military satellites</li> </ul> |
| Jun. 2023 | <ul style="list-style-type: none"> <li>• Hacking attack on the JAXA</li> <li>- Hacking of a JAXA server leading to theft of personnel data from internal networks</li> </ul>                                                                                                                                     |
| Oct. 2023 | <ul style="list-style-type: none"> <li>• Cyberattack in a satellite network of Korea</li> <li>- Attempted hacking of a private network through analysis of unencrypted satellite signals and unauthorized access to the satellite network</li> </ul>                                                             |

## Section 2      Scenarios of Cyberattacks on Space Systems

This document identifies threats of cyberattacks on space systems in different stages of a space system's life cycle, from design and development to operation and use, and to decommissioning, and outlines 15 probable scenarios of cyberattacks on space systems based on an analysis of major past cyberattacks.

For each scenario, the order and target of the attack and the estimated resulting loss are indicated, to be referenced in planning space system cybersecurity.

The cyberattack scenarios were categorized by space system components, i.e. ground station, satellite network (wireless or wired), and satellite.

### Scenarios of Cyberattacks on Ground Stations

- ① Email containing malware is sent, and the malware is installed in a space system connected to the internet, giving the attacker control access
- ② Distribution of an update file containing malware to hijack control of a space system
- ③ Connection of a portable storage device (USB drive) containing malware to a space system resulting in malware infection and system paralysis
- ④ Unauthorized control of a space system by an insider for data theft and disabling of functions
- ⑤ Inadequately managed networks or information security systems allowing an attacker to breach a space system to steal data or take control of the space system
- ⑥ Radio interference with a ground station for disruption of ground station-satellite communication



### Scenarios of Cyberattacks on Satellite Networks

- 7 Illegal collection and analysis of unencrypted control/data signals between a ground station and satellite for the theft of control commands and other critical data
- 8 Collection of telecommands between a ground station and satellite for a replay attack intended to disable the satellite
- 9 Connection by an insider to a space system through an unauthorized network (tethering, WiFi, etc.), resulting in an internet connection which is exploited as an opening for hacking attacks
- 10 Use of a reproduced satellite communication terminal (disguised as a normal terminal) by the attacker for infiltration of a network connected to a satellite communication service

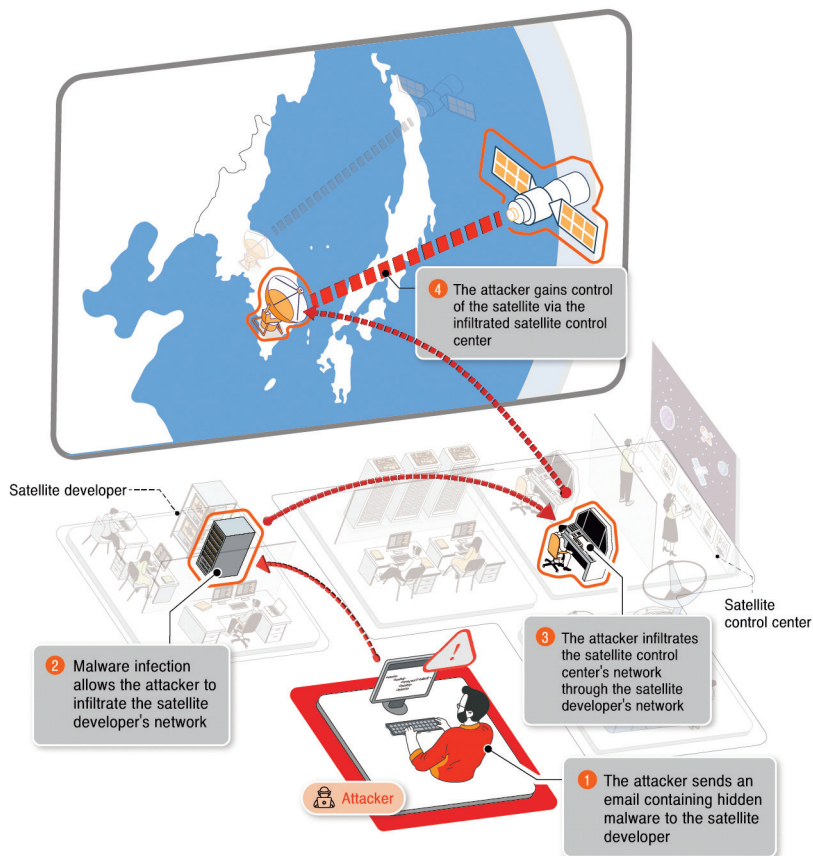
### Scenarios of Cyberattacks on Satellites

- 11 Exploitation of software security vulnerabilities in a space system (ground systems, satellites, etc.) for the paralysis or destruction of the space system
- 12 Exploitation of hardware security vulnerabilities in a space system (ground systems, satellites, etc.) for the paralysis or destruction of the space system
- 13 Transmission of unauthorized telecommands for manipulation of a satellite
- 14 Interference with the GNSS signals, leading to space system malfunctions
- 15 Theft of critical data or unauthorized control resulting from inadequate management of a mission-completed satellite

- 1 Email containing malware is sent, and the malware is installed in a space system connected to the internet, giving the attacker control access

The attacker sends an email embedded with malware to an employee with a developer/manufacturer and installs a “backdoor” (a program granting remote access), gaining access to a space system (ground station, satellite, etc.) connected to the internet.

Fig. 2-1 | Email attack with malware injection



## Attack Sequence

- ① The attacker sends an email embedded with malware to a person involved with a space system, e.g. employees of the development and manufacturing company
  - Based on researched data, the attacker gives the email a subject of interest to the recipient
- ② If the recipient opens the email containing malware, their computer becomes infected by the malware. The infected computer distributes the malware to other devices on the internal network or spreads the infection through a portable storage device, resulting in several computers with access to a space system becoming infected.
- ③ Infected computers give the attacker direct access to a ground station's satellite control system or spread the malware to the satellite control system through a dedicated line or VPN<sup>3)</sup>.
- ④ The infected satellite operation system accesses a satellite in operation through a satellite network to perform malicious controls or disable key functions.

## Attack target

- Personnel of entities involved in the development, manufacture, control, and use of space systems

## Expected damage

- Leaks of satellite operation data

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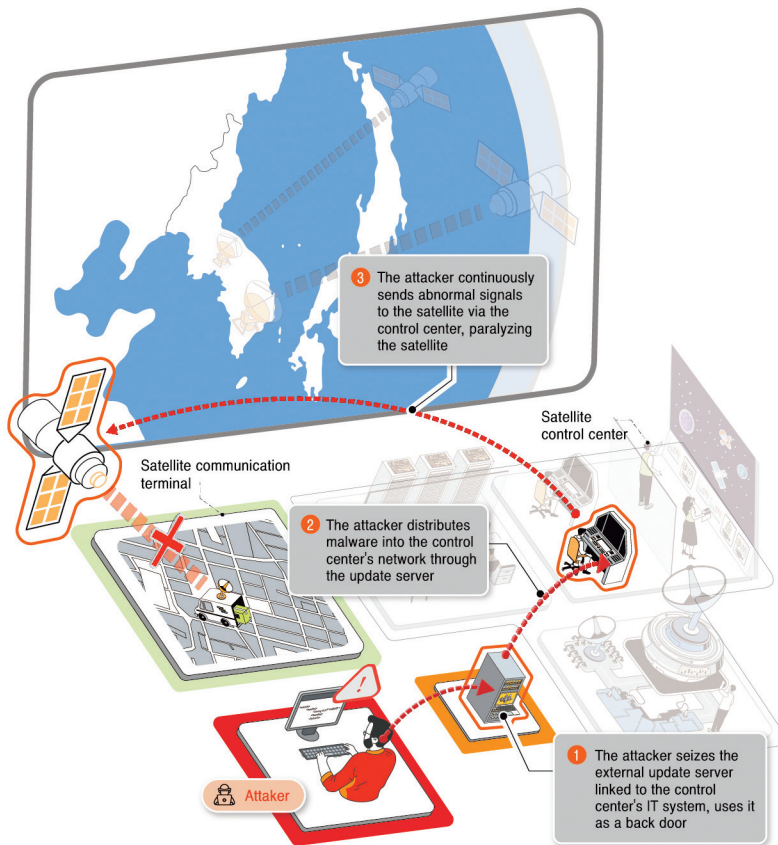
3) VPN : Virtual Private Network

- Configurations and operations (frequencies, orbits, etc.) of satellite systems
- Disruption and damage of satellite mission data
  - Satellite systems and their frequencies, orbits, mission data, etc.
  - Mission data reception failure
- Hijacking of satellite control
  - interference with normal operation
  - Disabling of communications and other functions

## 2 Distribution of an update file containing malware to hijack control of a space system

The attacker sends an update file embedded with malware to an update server connected to the internet to take control of the space system (ground station, satellite, etc.) where the update file is installed and steal data or disable functions.

Fig. 2-2 | Distribution of an Update File Containing Malware



## Attack Sequence

- ① The attacker hacks a developer, manufacturer, or maintenance service provider, or hacks an external update server connected to a ground station (satellite control center) and replaces a legitimate update file with one containing malware.
  - Instead of hacking an update server, the attacker can send an email with the download link to the malicious update file.
- ② Servers, network equipment, or communication terminals connected to a space system mistakes the malicious update file distributed through the update server for a normal update file. The malicious update file is downloaded and installed, infecting the space system.
- ③ Infected servers, network equipment, or communication terminals accesses other space systems through a satellite network to perform malicious controls or disable key functions.

## Attack target

- Update server of developers, manufacturers, or maintenance service providers

## Expected damage

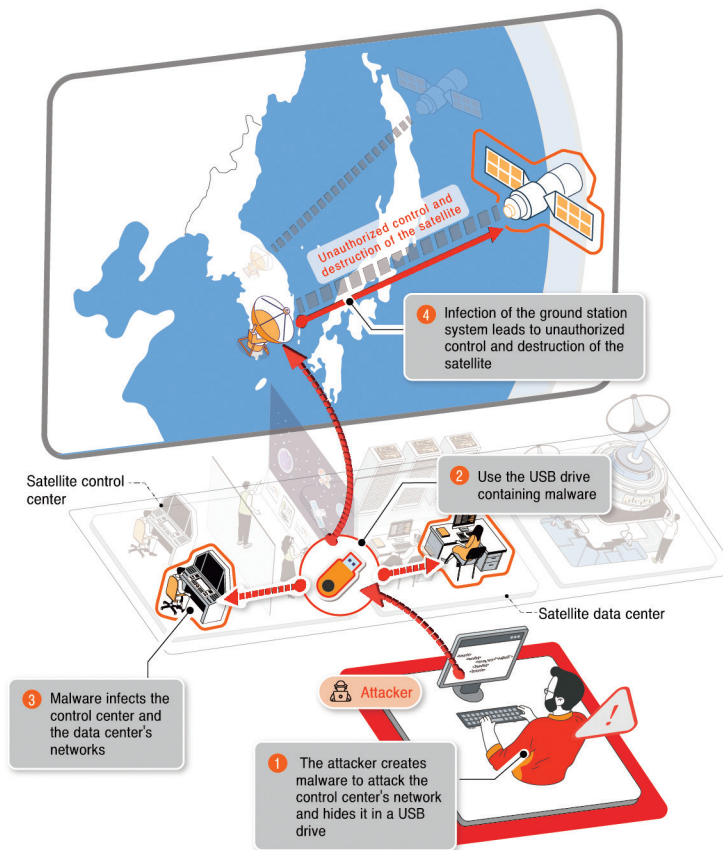
- Leaks of satellite control and operational data
  - Configurations and operations (frequencies, orbits, etc.) of satellite systems
- Hijacking of satellite control
  - Interference with normal operation

- Disabling of communication and other functions
- Paralysis of ground stations
  - Interference with normal operation
  - Malfunction, Disabling of functions

### 3 Connection of a portable storage device (USB drive) containing malware to a space system, resulting in malware infection and system paralysis

The attacker conceals malware for disabling a ground station system or taking control of a satellite in a portable storage device (USB drive) for use, targeting a person involved with a space system. If the portable storage device is connected to a ground station system, the malware gets installed and paralyzes the system.

Fig. 2-3 | Connection of a Portable Storage Device Containing Malware





## Attack Sequence

- ① Taking a route such as email, a malicious website, or a public PC, the attacker conceals malware in a portable storage device used by a person involved with a space system.
- ② The portable storage device is connected to a system at a ground station (satellite control center, satellite data center, etc.).
- ③ The malware hidden in the portable storage device is released into the ground station through computers or servers.
- ④ The malware gives the attacker access to a satellite in operation through a satellite network, resulting in unauthorized control or disabling of functions.

## Attack target

- Terminals of space system personnel, satellite control systems (servers)

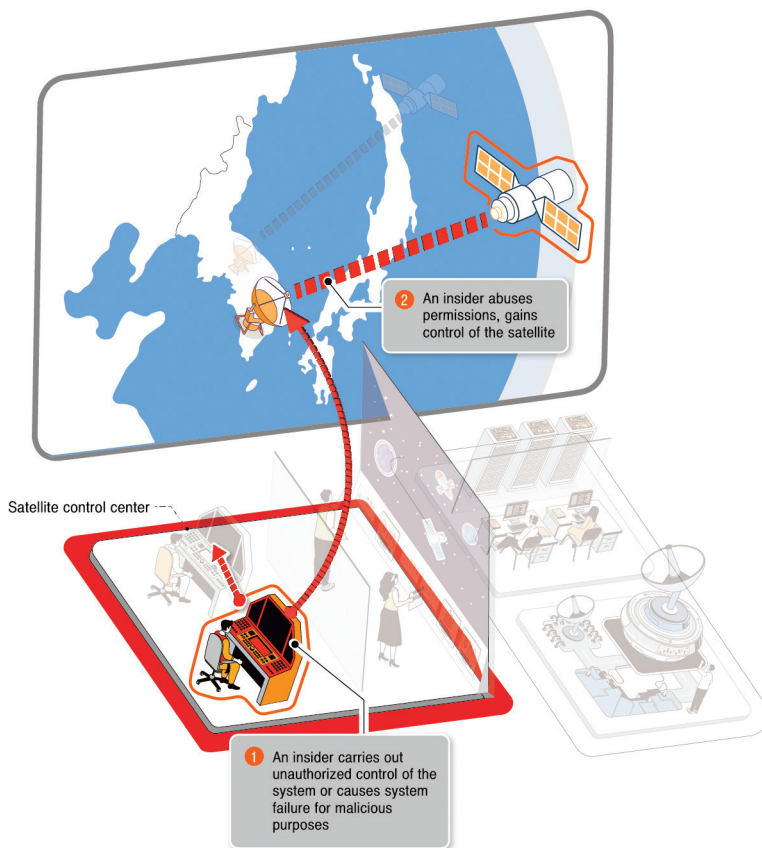
## Expected damage

- Leaks of satellite operation data
  - Configurations and operations (frequencies, orbits, etc.) of satellite systems
- Leaks of satellite mission data
  - Satellite systems and their frequencies, orbits, mission data, etc.
  - Mission data (high-resolution images, etc.)
- Hijacking of satellite control
  - interference with normal operation
  - Disabling of functions

#### 4 Unauthorized control of a space system by an insider for data theft and disabling of functions

An insider with a developer, manufacturer, or space system with malicious intentions can take control of a space system to steal data or disable certain functions.

Fig. 2-4 | Unauthorized Control of a Space System by an Insider



### Attack Sequence

- ① An insider with access permissions bypasses physical, administrative, or technical security measures to take control of a space system or disable certain functions.
- ② The insider uses access permissions to perform unauthorized control of a satellite.

### Attack target

- Computers, servers, data of developers and manufacturers
- Space systems of entities involved in the control and use of satellites

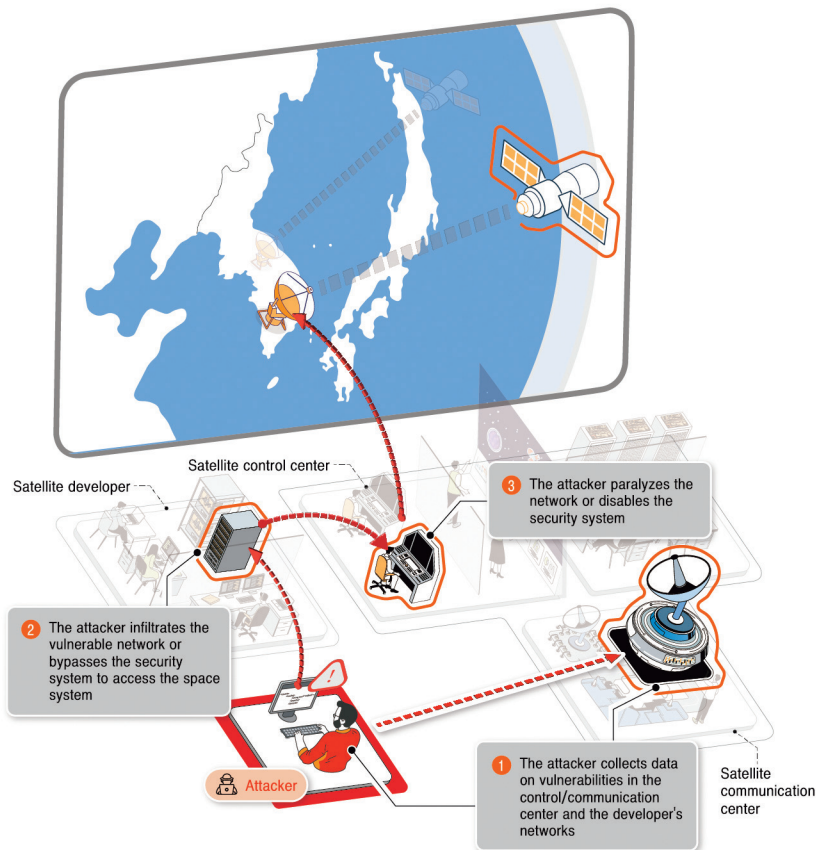
### Expected damage

- Leaks of satellite operation data
  - Configurations and operations (frequencies, orbits, etc.) of satellite systems
- Leaks of satellite mission data
  - Satellite systems and their frequencies, orbits, mission data, etc.
  - Mission data (high-resolution images, etc.)
- Hijacking of satellite control
  - Interference with normal operation
  - Disabling of functions

5 Inadequately managed networks or information security systems allowing an attacker to breach a space system to steal data or take control of the space system

Inadequate management of networks or information security systems making up a space system can lead to the destruction of the space system or data leaks.

Fig. 2-5 | Inadequate Management of a Network or Information Security System



## Attack Sequence

- ① The attacker uses network scanning or other means to collect data on networks and information security systems used by a space system or developer, such as routers, firewalls, intrusion prevention systems, and VPN equipment.
  - Collection of public data (history of system purchases, partner portfolios, etc.)
  - IP band scanning for identification of external network vulnerabilities
  - Internal network scanning through VPN permission theft or network interference
- ② Taking advantage of vulnerabilities in networks or information security systems without adequate security measures, the attacker breaches or bypasses a network or information security system to access the space system of its target.
  - Attacks on major networks or information security systems in a supply chain
  - Manipulation of known vulnerabilities in systems, e.g. inadequate security patches or configurations
  - Unauthorized connection through an exposed administrator's page or a remote connection service
- ③ Having infiltrated the network or information security system, the attacker paralyzes the network or information security system or makes configuration changes, and attempts additional attacks on other space systems or satellites.

## Attack target

- Networks and information security systems of entities involved in the development, manufacture, control, and operation and use of space systems

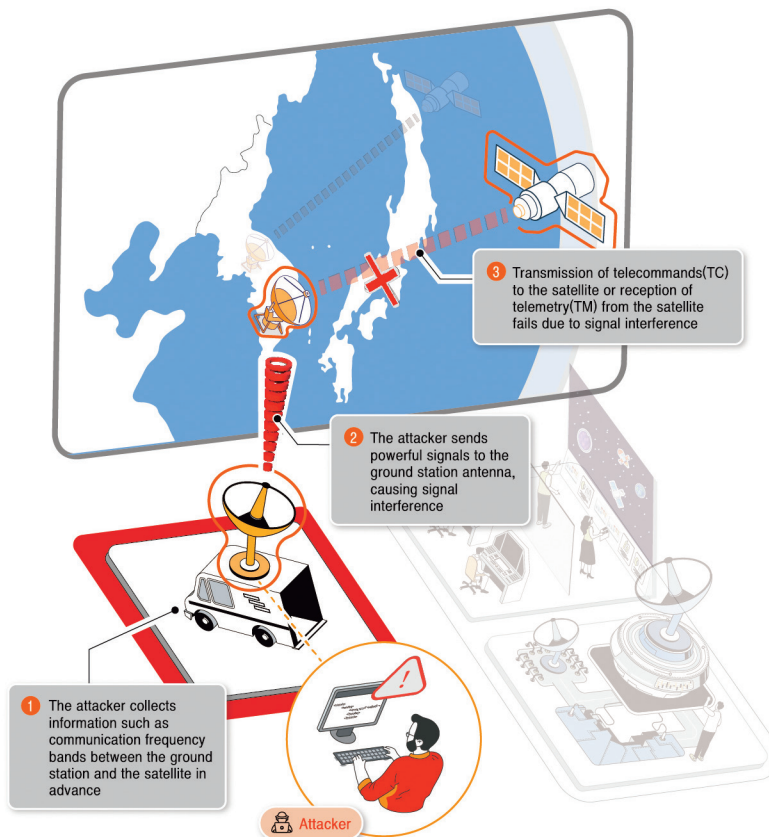
## Expected damage

- Network paralysis
- Information security system failure, leaving systems open to attacks
- Leaks of satellite operation data
  - Configurations and operations (frequencies, orbits, etc.) of satellite systems
- Leaks of satellite mission data
  - Satellite systems and their frequencies, orbits, mission data, etc.
  - Mission data (high-resolution images, etc.)
- Hijacking of satellite control
  - Interference with normal operation or disabling of functions

## 6 Radio interference with a ground station for disruption of ground station-satellite communications

The attacker sends powerful radio interference signals to a ground station antenna to disrupt communications between the ground station and a satellite, preventing the ground station from sending telecommands(TC) or receiving status telemetries(TM) from satellites.

Fig. 2-6 | Radio Interference with a Ground Station



## Attack Sequence

- ① Based on researched data, the attacker estimates the ground station-satellite communication frequency bands and ground station antenna operating times.
  - Public data available to attackers includes business reports or research papers posted on the internet. Data can also be collected from radio signals.
  - The attacker can visually identify a ground station's antenna operating times.
- ② Using the same frequency band as a ground station antenna, the attacker generates powerful radio signals to interfere with radio signals received by the ground station.
- ③ The attacker's interference with radio signals prevents a satellite from receiving telecommands from the ground station or prevents the ground station from receiving status telemetry (mission data) from a satellite.

## Attack target

- Ground station antennas or radio communications between ground stations and satellites

## Expected damage

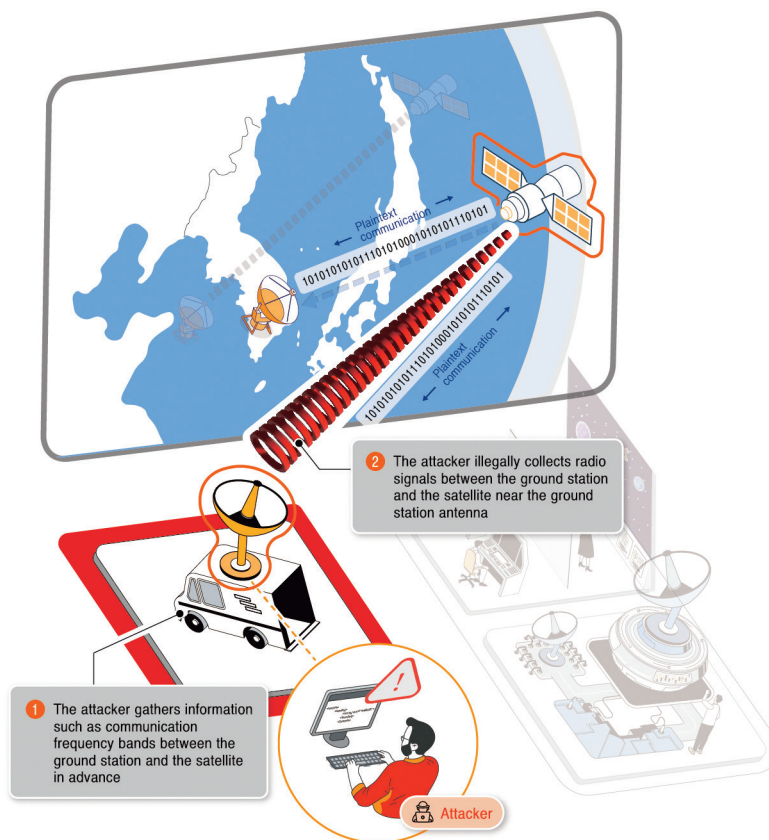
- Disruption of communications with satellites
  - Failure of ground stations to send telecommands to satellites
  - Failure of ground stations to receive telemetry from satellites



## 7 Illegal collection and analysis of unencrypted control/data signals between a ground station and satellite for the theft of control commands and other critical data

An attacker can illegally collect and analyze unencrypted control/data signals between a ground station and satellite to gain critical data.

Fig. 2-7 | **Illegal Collection and Analysis of unencrypted Signals Between a Ground Station and Satellite**



## Attack Sequence

- ① Based on researched data, the attacker estimates ground station-satellite communication frequency bands and ground station antenna operating times.
  - Public data available to attackers includes business reports or research papers posted on the internet. Data can also be collected from radio signals.
  - The attacker can visually identify a ground station's antenna operating times.
- ② The attacker illegally collects unencrypted radio signals in satellite frequency bands to and from a ground station from near the ground station's antenna. Analyzing the collected signals, the attacker understands ground station-satellite or Inter-satellite links.
  - The attacker can take advantage of multiple satellites in space to collect radio signals.

## Attack target

- Unencrypted radio signals on satellite networks (telecommands, telemetry, mission data, etc.)

## Expected damage

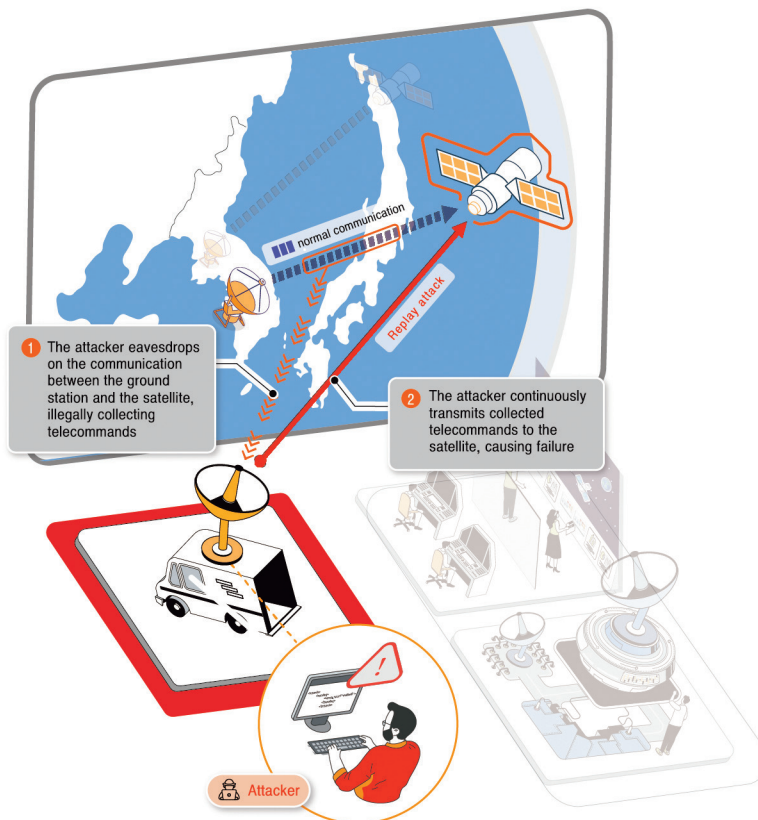
- Leaks of satellite operation data
  - Configurations and operations (frequencies, orbits, etc.) of satellite systems
- Leaks of satellite mission data
  - Leaks of satellite TC and TM formats

- Generating new TC commands from the TC format to disrupt the satellite's mission or, in worst case, seize control of the satellite
- Analyzing the TM format to obtain the satellite's status and assess the success of TC attacks

## 8 Collection of telecommands between a ground station and satellite for a replay attack intended to disable the satellite

The attacker collects telecommands transmitted between a ground station and satellite, and executes a replay attack to disable or destroy the satellite.

Fig. 2-8 | **Replay Attack Using Telecommands Exchanged Between a Ground Station and Satellite**



## Attack Sequence

- ① The attacker illegally collects telecommands in satellite frequency bands to and from a ground station from near the ground station's antenna.
  - Based on researched data, the attacker estimates ground station-satellite communication frequency bands and ground station antenna operating times.
  - Public data available to attackers includes business reports or research papers posted on the internet. Data can also be collected from radio signals.
  - The attacker can visually identify a ground station's antenna operating times.
  - The attacker can take advantage of multiple satellites in space to collect radio signals.
- ② The attacker sends collected signals back to the satellite to disable the satellite.

## Attack target

- Satellite

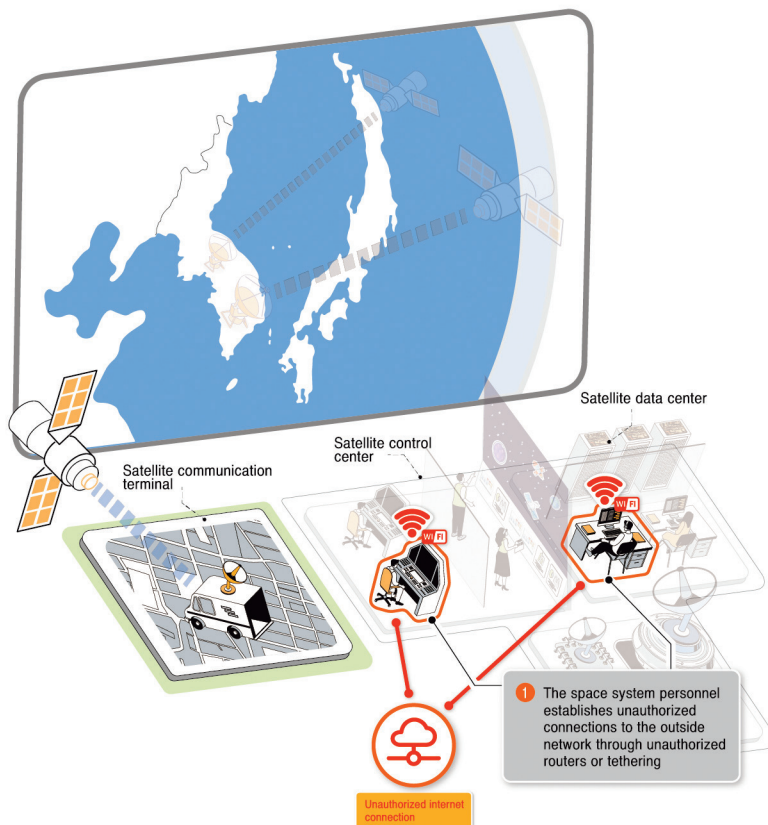
## Expected damage

- Hijacking of satellite control
  - Interference with normal operation
  - Disabling of functions

**9** Connection by an insider to a space system through an unauthorized network (tethering, WiFi, etc.), resulting in an internet connection which is exploited as an opening for hacking attacks

A person involved with a space system uses a satellite control system at a ground station to connect to an unauthorized network by a wired or wireless LAN card or by smartphone tethering, creating an internet connection which is exploited as an opening for hacking attacks.

**Fig. 2-9 | Connection of a Space System to an Unauthorized Network**



## Attack Sequence

- ① The person connects an unauthorized network to the satellite control network.
  - Connection of a wired or wireless router or smartphone to a satellite control system at a ground station (connection to an outside commercial network), creating an internet connection between the satellite control network and outside commercial network
  - Exploitation of the connection between the satellite control network and external commercial network by the attacker for malware distribution, or exploitation of vulnerabilities in the satellite control system, can lead to the seizure of a space system
  - The attacker uses the seized space system to steal data (control data, satellite mission data, etc.), or disables the space system

## Attack target

- Neglected, unauthorized networks

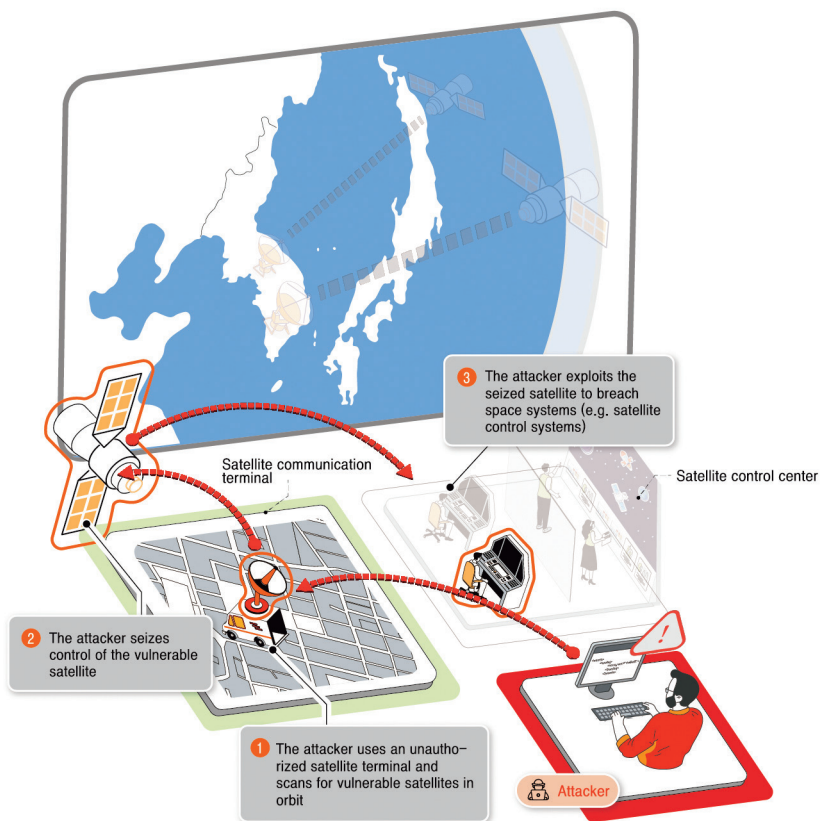
## Expected damage

- Leaks of satellite operation data
  - Configurations and operations (frequencies, orbits, etc.) of satellite systems
- Leaks of satellite mission data
  - Satellite systems and their frequencies, orbits, mission data, etc.
  - Mission data (high-resolution images, etc.)
- Hijacking of satellite control
  - Interference with normal operation
  - Disabling of functions

**10** Use of a reproduced satellite communication terminal (disguised as a normal terminal) by the attacker for infiltration of a network connected to a satellite communication service

The attacker can use a reproduced satellite communication terminal to breach through a satellite a network connected to satellite communication services.

**Fig. 2-10 | Use of a Reproduced Satellite Communication Terminal (Disguised as a Normal Device) for Internal Network Infiltration**





## Attack Sequence

- ① The attacker uses a stolen satellite communication terminal or a reproduced one to scan for satellites in orbit vulnerable to Hijacking or manipulation.
  - The attacker can build a satellite communication terminal by referring to publicly available information such as reports and papers, or purchase and modify a similar device.
- ② The attacker uses the reproduced satellite communication terminal to attack and seize the compromised satellite with inadequate security, or achieves normal communication.
- ③ The attacker breaches and seizes a space system by disabling a satellite communication service or through a satellite communication service.
  - The attacker reverse engineers or modifies the communication terminal as needed to carry out attacks.

## Attack target

- Satellite network (wired or wireless)

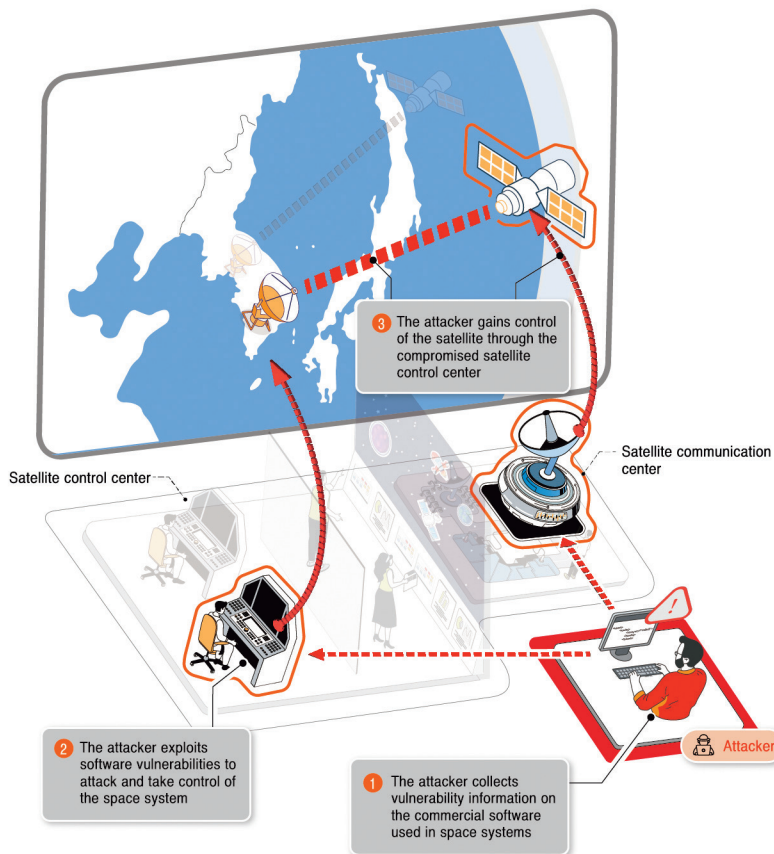
## Expected damage

- Disabling of satellite communication services
  - Malfunction of satellite communication services
  - Denial of satellite communication services
- Infiltration of connected networks or systems

## 11 Exploitation of software security vulnerabilities in a space system (ground systems, satellites, etc.) for the paralysis or destruction of the space system

The attacker can exploit security vulnerabilities in software configured in a space system (ground stations, satellites, etc.) to paralyze or destroy the space system.

Fig. 2-11 | Exploitation of Software Security Vulnerabilities in a Space System



## Attack Sequence

- ① The attacker collects data on software a space system (ground stations, satellites, etc.) uses, and analyzes its security vulnerabilities.
  - The attacker can research publicly available information such as reports and papers or physically obtain a space system (satellite communication terminal) or reverse engineer one to collect data on software used in system development (open sources, operating systems, programs, etc.).
- ② The attacker creates malware to exploit software vulnerabilities and uses it to attack and seize a space system.
  - Outdated software can be more vulnerable to advanced hacking techniques.
  - Use of operating systems or programs that have not been updated for various reasons (compatibility issues, etc.) increases the likelihood of software security vulnerabilities being exploited, potentially leading to the paralysis or destruction of space systems.
  - Vulnerabilities in open-source software being used can undermine the security of an entire system.
  - Software updates can result in incompatibility with an operating system or software, causing functions normal hitherto to malfunction.
- ③ The attacker uses a seized space system to gain control of a satellite.

## Attack target

- Space systems (satellites, ground stations, etc.)

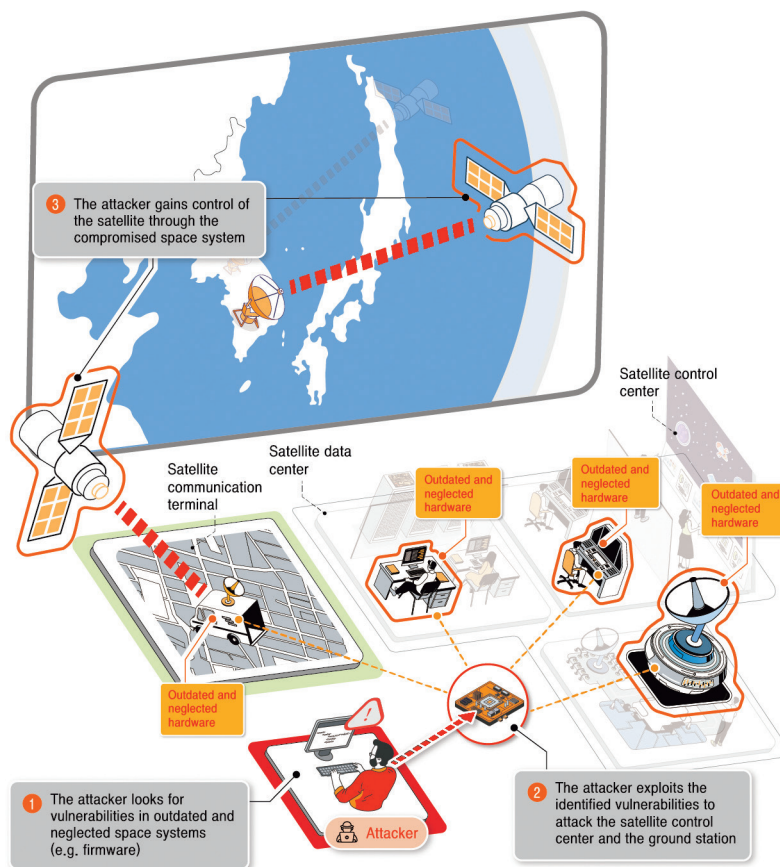
## Expected damage

- Paralysis of ground stations
  - Interference with normal operation
  - Disabling of functions

## 12 Exploitation of hardware security vulnerabilities in a space system (ground systems, satellites, etc.) for the paralysis or destruction of the space system

The attacker can exploit vulnerabilities in outdated hardware (firmware) of a space system (ground stations, satellites, etc.) to paralyze or destroy the space system.

Fig. 2-12 | Exploitation of Hardware Security Vulnerabilities in a Space System



## Attack Sequence

- ① The attacker collects data on hardware configured in a space system, and analyzes its vulnerabilities.
  - The attacker can use publicly available information such as reports and papers or steal and dismantle a space system (satellite communication terminal) to collect hardware data.
- ② The attacker exploits hardware vulnerabilities to attack a space system (ground systems, satellites, etc.).
  - Outdated hardware can be more vulnerable to advanced hacking techniques.
  - Use of outdated hardware components on account of compatibility creates the risk of attacks targeting hardware vulnerabilities to paralyze or destroy space systems.
- ③ The attacker exploits the compromised space system to gain control of a satellite.

## Attack target

- Space systems (satellites, ground stations, etc.)

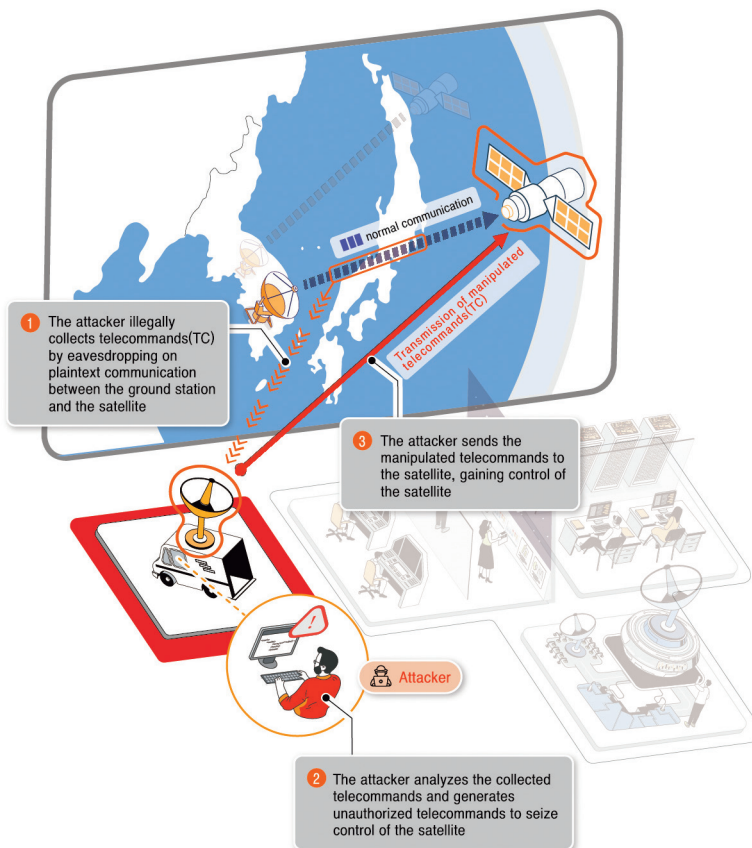
## Expected damage

- Hijacking of satellite control
  - Interference with normal operation
  - Disabling of functions
- Paralysis of ground stations
  - Interference with normal operation
  - Disabling of functions

### 13 Transmission of unauthorized telecommands for manipulation of a satellite

An attacker can collect and analyze telecommands transmitted between a ground station and satellite to create and transmit telecommands (TC) capable of controlling the satellite, potentially manipulating or destroying the satellite.

Fig. 2-13 | Transmission of Unauthorized Telecommands to a Satellite by an Attacker



## Attack Sequence

- ① The attacker illegally collects wireless signals in satellite frequency bands to and from a ground station from near the ground station's antenna.
  - Based on researched data, the attacker estimates ground station-satellite communication frequency bands and ground station antenna operating times.
  - Public data available to attackers includes business reports or research papers posted on the internet. Data can also be collected from wireless signals.
  - The attacker can visually identify a ground station's antenna operating times.
  - The attacker can take advantage of multiple satellites in space to collect wireless signals.
- ② The attacker analyzes the collected telecommands to identify protocols of transmission between a ground station and satellite or between satellites, and creates telecommands to gain control of a satellite.
- ③ Using the identified telecommand transmission protocols, the attacker directly transmits manipulated satellite telecommands to a satellite and gains control of the satellite.

## Attack target

- Satellites

## Expected damage

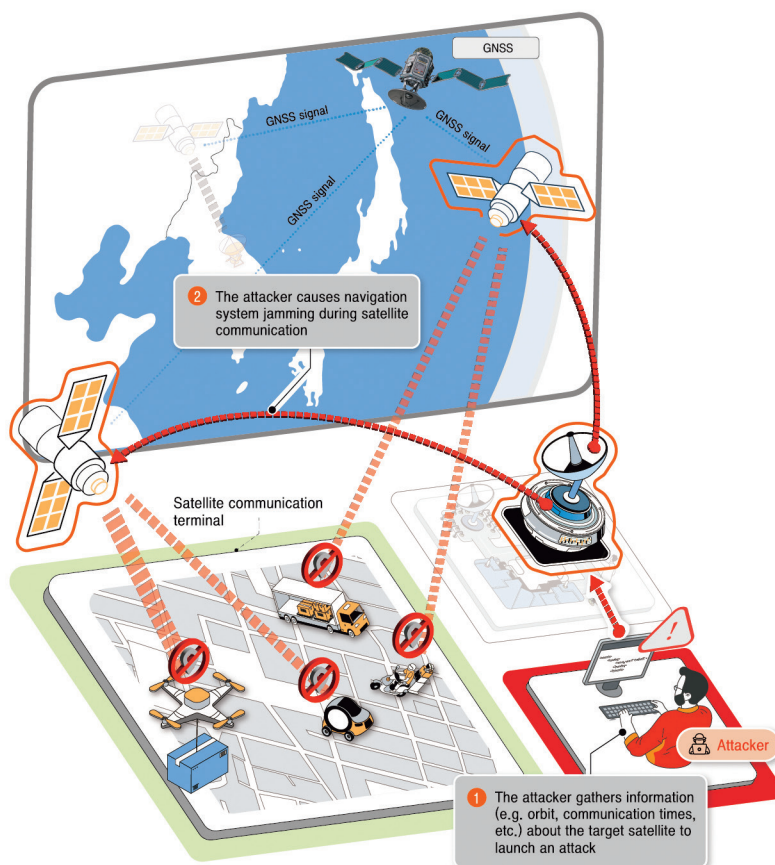
- Hijacking of satellite control
  - Interference with normal operation
  - Disabling of functions



## 14 Interference with the Global Navigation Satellite System signals, leading to space systems malfunctions

The attacker can interfere the GNSS signals, leading to space systems malfunctions.

Fig. 2-14 | Disruption of a Satellite Navigation System Used by a Space System



## Attack Sequence

- ① The attacker collects data on the orbit, communication times, and other conditions of a satellite to attack.
  - Public data available to attackers includes business reports or research papers posted on the internet. Data can also be collected from radio signals.
- ② The attacker generates signals to disrupt the satellite navigation system to coincide with the targeted satellite's communication. Communication of location, time, or other conditions is disrupted, causing the satellite to malfunction or become unable to execute its mission.

## Attack target

- Satellites, ground stations, etc.

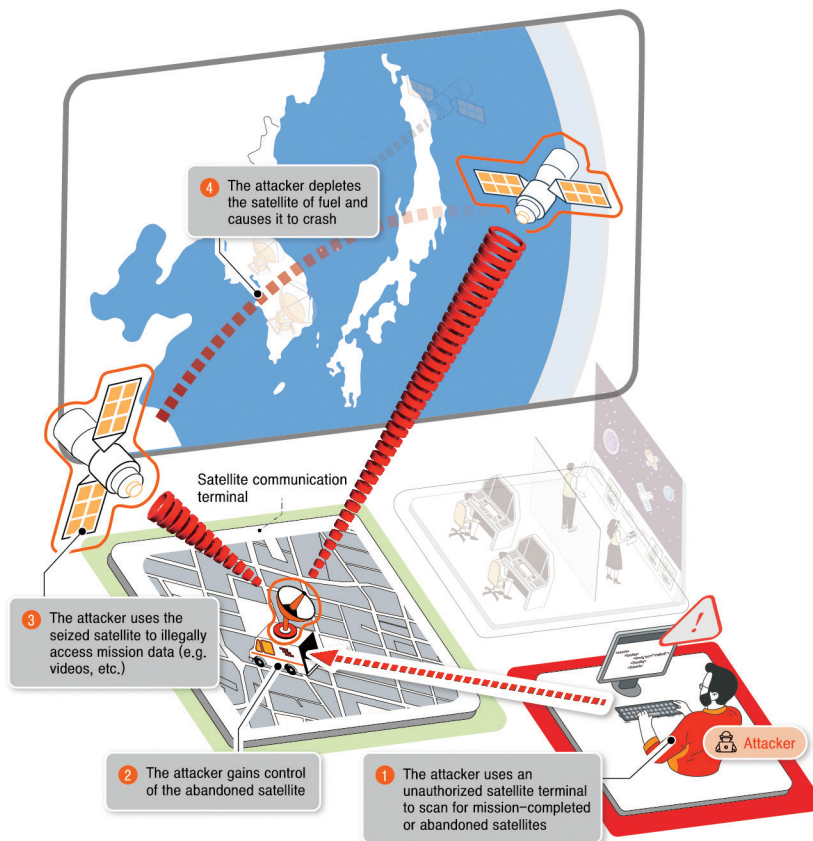
## Expected damage

- Failure of ground station-satellite communications
- Failure of satellite communication terminals (satellite communication services)

**15** Theft of critical data or unauthorized control resulting from inadequate management of a mission-completed satellite

Inadequate management such as leaving a mission-completed satellite idle can create an opening for attackers to steal data or gain control of the satellite to cause loss.

**Fig. 2-15 | Attack Resulting from Inadequate Management of a Mission-completed Satellite Order of Attack**



## Attack Sequence

- ① The attacker uses an unauthorized satellite terminal to scan for mission-completed or abandoned satellites in space.
  - Satellites left in space, intact and still in orbit
- ② The attacker attacks an identified satellite and gains control of it.<sup>4)</sup>
  - Gaining of control through the exploitation of hardware or software security vulnerabilities
- ③ The attacker steals mission data (images, etc.) stored in the satellite or controls the satellite to film certain locations.
- ④ The attacker controls the stolen satellite to cause a collision with another satellite by depleting its fuel.

## Attack target

- Mission-completed satellites, ground stations, etc.

## Expected damage

- Leaks of satellite operation data
  - Configurations and operations (frequencies, orbits, etc.) of satellite systems
  - Use of stolen data in further attacks on satellites of the same country or manufacturer

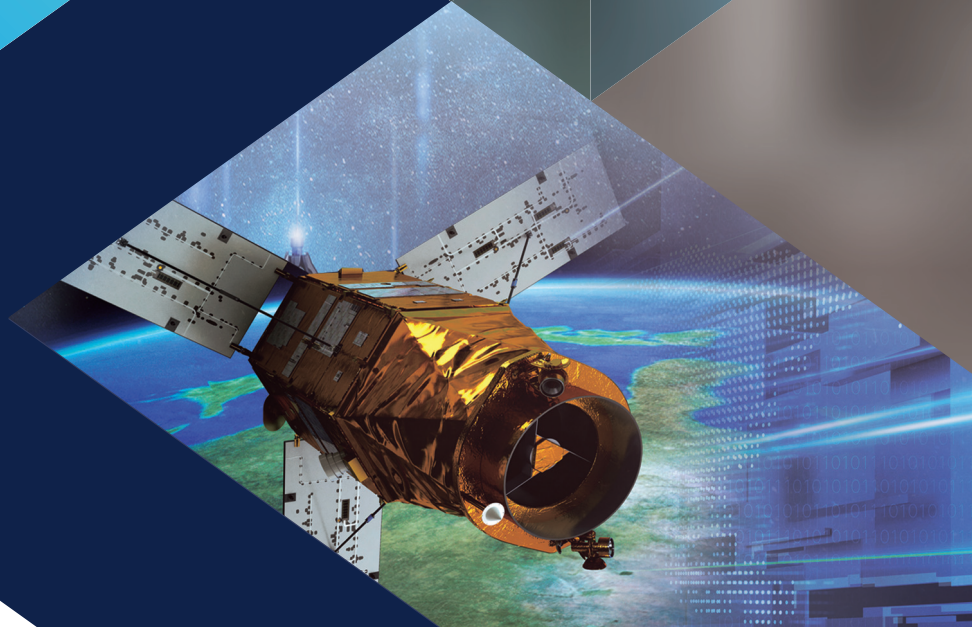
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4) A group of hackers gained control of an idle satellite at an international hacking competition in 2022.

- Leaks of satellite mission data
  - Mission data (high-resolution images, etc.)
- Hijacking of satellite control
  - Deorbiting of a satellite to cause a collision with another satellite

## Chapter 3

# Cybersecurity measures for Space Systems







## Chapter 3

# Cybersecurity measures for Space Systems

In this part of the document, security measures to be adopted for space systems such as ground stations, satellite networks (on wired or wireless networks), and satellites are described based on [Fig. 1-1] Standard Model for Space System presented in Section 3 of Part 1 and the 15 scenarios of cyberattacks on space systems described in Section 2 of Part 2.

As the application of new security measures to a space system starting or already in operation and use faces several challenges, security measures of an administrative and technical nature should be applied in advance throughout the entire space system lifecycle, starting from the design and development of space systems.

However, this document does not propose security measures against threats involving electromagnetic pulses(EMP). Such security measures can be developed by referencing the guidelines relevant to a given country.



## Section 1

# Cybersecurity measures in the design and development phase

## Administrative Security Measures (Applicable to All Space Systems)

As the production of information concerning space systems starts in their design and development, administrative security measures against cyber threats should be prepared and adopted by operators of space systems and all outsourcing companies involved.

### 1. Planning of the Cybersecurity of Space Systems

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From the time of the design and development of a space system the implementation of cybersecurity across its entire life cycle should be planned based on its purpose and development.

- Organization for the Cybersecurity of Space Systems
- Designation and management of protected areas
- Management of outsourcing companies
- Categorization and management of critical data
- Hardware and software development security management
- Decommissioning of space system-related devices and data

## 2. Designation and Management of Protected Areas

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To achieve high levels of security in the design and development of ground stations (satellite control centers, satellite data centers, and satellite communication centers) and satellites, locations requiring physical protection are to be designated as protected areas and controlled in physical access.

### Control of Outside Information and Communications Devices

- The presence of outside information and communications devices (laptop computers, portable storage devices, etc.) in protected areas is to be controlled.
  - Installation of only programs needed for work as approved by the CISO (chief information security officer)
  - Advance scanning of information and communications devices brought into protected areas for malware and unnecessary programs
  - Logging of information and communications devices brought into and removed from protected areas
  - No connection of outside information and communications devices to a space system in development
  - Installation and use of necessary outside programs only on specified laptop computers or portable storage devices following all due security measures, including CISO's approval and scanning for malware

## Authorization and Identification of Persons on Entry/Exit

- Operation of 24/7 security-monitored entries and exits capable of the authorization and identification of persons entering and exiting to limit the access of protected areas to persons involved in the design and development of space systems only
  - Biometric devices (fingerprint recognition) are recommended for personnel authorization and identification, or ID cards can be used as an alternative
  - Operation of a single main point of entry/exit (the one always used) equipped with personnel authorization/identification devices

## Prevention of Data Leaks

- Prevention of data leaks from protected areas by the planning of secure data export and emergency deletion, and the prevention of photography / filming

## 3. Security Management for Outsourcing Companies

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The use of outsourcing companies in the design and development of a space system is to be preceded by security clearance of the participating persons and equipment, and the formation of product security measures. Control measures are to be implemented around space systems accessible by outsourcing companies to prevent alterations and data theft.

### Control of Outsourcing Companies' Work Areas

- In-house setup of outsourcing companies' work areas to observe all due security measures

- Separation of information and communications networks used in outsourcing companies' work areas from the space system operator's information and communications networks
  - In instances requiring the access of the space system operator's information systems by outsourcing companies, the grant of limited access to only the necessary information systems from only the designated terminals, combined with access logging
- No unauthorized information and communications devices allowed in outsourcing companies' work areas; authorized devices to be port-sealed and security-stickered

### Control of Outsourcing Companies' Remote Work

- No development or maintenance activities from remote locations allowed; in unavoidable instances, the CISO to issue approval and implement due security measures beforehand
  - Permission of connection from designated terminals only, with the designated terminals allowed connection to only the necessary development systems through the space system operator's outsourcing company online security control system
  - No use of designated terminals for unauthorized purposes; operating systems to be kept updated; no installation of unnecessary programs
  - Encryption of communication in remote work; activation of communication only during work hours, followed by immediate deactivation
  - The space system operator to issue advance approval of the particulars and duration of remote work to take place

- And history of remote work to be monitored by the space system operator's personnel

## Equipment Security Management

- The space system operator to provide and log equipment to be used by outsourcing companies
- In instances requiring the use of a outsourcing company's own equipment, the space system operator to take all due security steps (inspection and reset of equipment) before approving and logging the use
  - Equipment with components banned from use (wireless LAN equipment) on site to be identified and quarantined
- Physical sealing of ports and installation of security systems to prevent the use of unauthorized storage devices and communications devices

## Security Training and Inspection of Outsourcing Companies

- Obtaining of security agreements from outsourcing companies, training of outsourcing companies in security protocols, and regular and unscheduled, frequent inspection of security compliance for the identification and correction of issues
  - No copying or release to the outside of the space system operator's data
  - No work-related equipment to be brought onto or removed from work sites without authorization
  - No connection to unauthorized networks from smartphones

## 4. Management of Space System Developer Accounts

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Space system developer accounts are to be created only for persons who require one for work, with all due security measures implemented.

- Creation of an account for approved persons only; only one account person (no account sharing)
- Created accounts to provide essential permissions only; permissions to be regularly reviewed and updated
  - Immediate deletion or change of accounts in keeping with changes in personnel (transfers, exits, change of duties, etc.)
- All accounts to be set up with a password, a combination of numbers, letters, and symbols difficult to guess and regularly changed

## 5. Safeguarding of Critical Data

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Critical data concerning space systems, such as dimensions, specifications, control signal protocols, and frequency modulation, is to be safeguarded as its exposure to outside entities creates the risk of cyberattacks.

- No upload of research or supplier data concerning space systems to internal or external public servers
  - Critical data uploaded to public servers to be redacted or deleted
  - Encryption of documents containing critical data for storage and access control
- Control of products of space system development made public (papers, reports, patents, etc.) to exclude critical data

## 6. Supply Chain Security Management (Hardware and Software Included)

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In the design and development of space system, hardware and software security management should be performed in the development, distribution, adoption, and operation of hardware and software. Hardware and software parts suppliers should meet security requirements, and safe partnerships should be pursued to use only verified parts, which can include the avoidance of trade with businesses under sanctions by international organizations such as the United Nations.

Software development should involve secure coding, with vulnerabilities minimized through the identification, analysis, and management of vulnerabilities.

- Hardware and security suppliers to meet security requirements
- Identification of security requirements ahead of hardware and software design, to be applied to design, development, and testing
- Software development to entail secure coding measures, including input data validation, security of encryption keys and passwords, correction of errors, session control, and time and status management
- Identification, analysis, and management of vulnerabilities
- An SBOM<sup>5)</sup> to be supplied with open-source or third-party software used to indicate information such as supplier name, software components, and software version to be used in the management of vulnerabilities and the formation of risk management strategies (evasion, mitigation, acceptance, etc.)
- A contract is to be established to stipulate the management of hardware or software security vulnerabilities in space systems. The contract should prohibit

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5) SBOM : Software Bill of Materials

activities creating or ignoring such vulnerabilities, such as unauthorized connections and the installation of unauthorized data collection programs. Contractual parties must redress security vulnerabilities occurring in contract periods, warranty periods included, by hardware or software replacement or other solutions.

## 7. Identification and Management of Space System Assets

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From the time of design and development hardware and software assets used in space systems are to be identified and managed. Movable assets such as communication terminals are susceptible to theft, loss, and damage so should be regularly checked.

## 8. Space System Cybersecurity Crisis Response System

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Response to a cybersecurity crisis occurring in the design and development of space systems is to take place through close cooperation among the appropriate entities and in accordance with the crisis response procedures in place in each country.

- Working-level response to a cybersecurity crisis is to take place in the early stages of the crisis occurrence and include the isolation of the attacked network, removal of security vulnerabilities, and deletion or protection of the appropriate data
  - Notification to relevant authorities on internal findings
  - Recovery from failures through backup and other measures



- Coordinate response to take place: Update on developments upward through the ranks of authority, emergency communication network activation, alerts on prevention and response measures

## 9. Cybersecurity Training of Space System Personnel

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Cybersecurity training of space system developers is to be planned and executed at least once a year to promote awareness, preparedness, knowledge, and expertise in cybersecurity. Expert training providers should be used as necessary.

### Technical Security Measures

Safe operation of space systems requires a competent review and implementation of technical security measures from the time of the design and development of a space system. Advance tests should take place to validate that the implemented security measures will fulfill their functions in space.

## 1. Cybersecurity Measures for Ground Stations

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### Control of Administrator Permissions

- A function for the creation and operation of system administrator accounts
- Separation of administrator services and user services

- Access of administrator services via set IP addresses only
- Multifactor authentication, login/session control, and other security functions for administrator services

### Updates and Security Patches

- Updates and security patches for ground station systems from the time of design and development to minimize security vulnerabilities
- Use of network integration solutions for one-way transmission of update and patch files to ground station systems
  - Development of systems on another physical network, e.g. the network of a satellite control center, to involve the use of a clean computer for inspecting update files and transferring them in a dedicated portable storage device
- Operation of a separate test system for the validation of update/patch file security to minimize system errors and failures
- Development of a system with no compatible or available security patches to involve the development and implementation of special security measures in the system's development
  - Advance identification of cyber threats associated with the absence of security patches
  - Additional security measures, including a regular check of logs to identify issues

### Control of Unnecessary Services and Functions

- Programs and services not required for ground station operation to be disabled or removed by checking influence on ground station security in development stages

- Minimum permissions for necessary programs and services
- Completion of design and development to be followed by the disabling of external interfaces (debugging ports, etc.) and files and databases used only in development activities

## Detection and Removal of Malware

- Installation of vaccine programs on servers, PCs, and other IT infrastructure used in design and development, regular update of vaccine programs for a regular scanning for security hazards
- Transmission of outside data to a ground station to be preceded using network integration solutions and clean computers for scanning such data for malware
- Special security measures in instances where vaccine programs cannot be installed or updated
  - Advance identification of cyber threats associated with the absence of vaccine programs
  - Logging and regular inspection of active processes
  - Additional security measures, including a regular check of logs to identify issues

## Blocking of Connection of Unauthorized Storage Devices

- Prevention of the use of unauthorized portable storage devices (USB drives) by the physical blocking of ports or the use of information security solutions (device control systems)
- No auto opening of portable storage devices; logging of portable storage device use

## Enhanced Email Security

- Computers and servers connected to a ground station requiring emailing to be set up with email security solutions
- Encryption of emails and attachments

## Encryption of Critical Data

- Encryption (approved encryption modules) of control commands, mission data, and other critical data for storage and transmission

## Backup of Settings and Stored Data

- Backup of settings and stored data of equipment used at ground stations for operational stability and prompt recovery

## Log Management

- Creation and management of logs for efficiency of ground station operations and the management of incidents
- Preservation of logs for a minimum of one year (access restricted to administrators)
  - Types and times of activity: Log-on/-off, viewing/printing of data, etc.
  - Types and times of activity: Log-on/-off, viewing/printing of data, etc.
  - Activity results: Successful/failed connections, etc.

## Reception of Multiple Satellite Navigation Systems

- As ground station malfunction can result from satellite navigation system jamming, location and time data to be received from multiple reliable satellite navigation systems, such as GPS and Galileo

## 2. Cybersecurity Measures for Satellite Networks

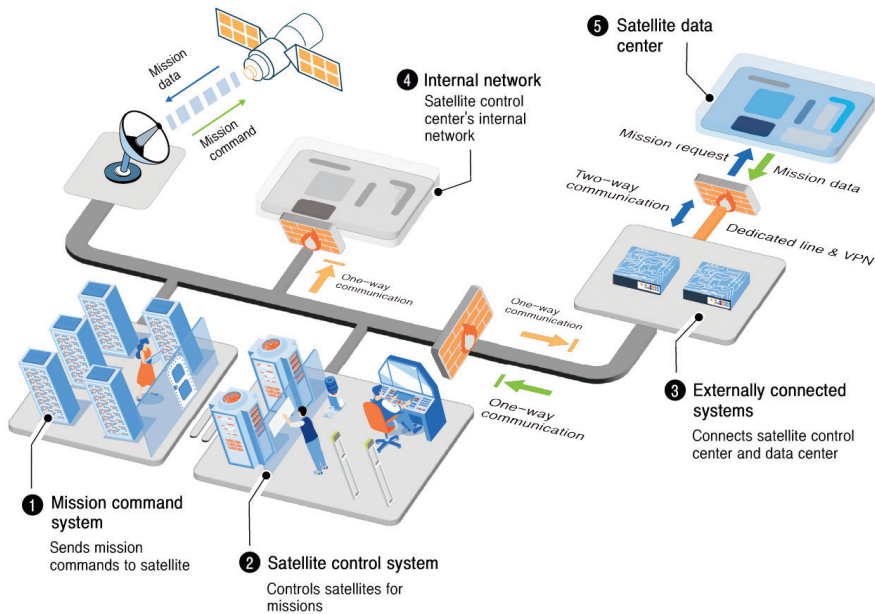
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### Ground Station Network Configuration

#### ① Satellite Control Centers

- Satellite control centers require cybersecurity measures commensurate with their role in the control of satellites, e.g. operation of satellites, mission planning, orbit determination
- Satellite control center networks not requiring transfers of data (mission and orbit data, etc.) to an internal network to be physically separated from internal or external networks
- Satellite control center networks to be configured as a physically independent, closed network; data to be transferred by one-way communication in instances where connection to an internal or external network is required for satellite operation data access (Fig. 3-1)

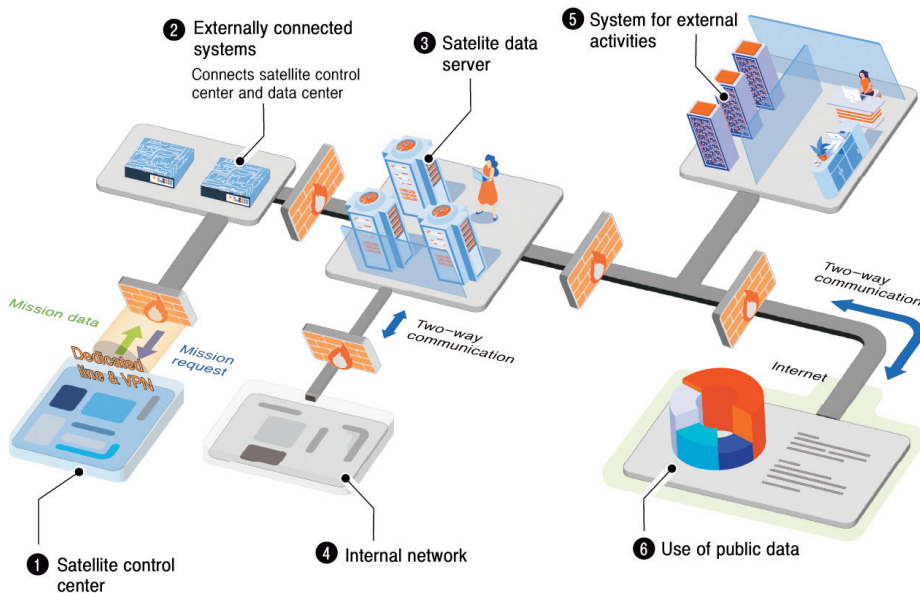
Fig. 3-1 | Satellite Control Center Network Configuration



## ② Satellite Data Centers

- Satellite data centers to be subject to strict control in external connections in view of their function of data collection through satellite control centers
- Collection of outside data from reliable sources only; use of network integration solutions for scanning for malware before internal transmission of data (Fig. 3-2)

Fig. 3-2 | Satellite Data Center Network Configuration



### ③ Satellite Communication Centers

- Strict control of unauthorized connections between satellite networks and internal networks with the use of security solutions such as network integration solutions and firewalls
- Separation of satellite networks in line with purpose, e.g. work, internet connection

### Ground Station Network Integration

- A ground station network requiring connection to a space system of another entity to be connected by a dedicated line, with connection by a VPN being an alternative

- Encryption of all sent and received data
- A ground station system to be connected to another space system to be separated to form a demilitarized zone (DMZ); information security solutions such as firewalls and anti-breach systems to be installed between the DMZ and ground station system
- Installation of network integration solutions in connections
- Ground stations equipped with cybersecurity measures suitable for a control system (satellite control centers) to be connected by one-way communication (Fig. 3-1)
  - One-way network for sending and receiving data to be separately set up only in instances where data transmission to a satellite control network is required
  - A DMZ to be set up before connecting a satellite control network to an external network, with the connection being established by one-way outbound communication and one-way inbound communication
  - Systems requiring external connection (connection to an outside entity) for data transmission to be connected by two-way communication through a dedicated line or VPN

### Encrypted Communication Configuration

- Satellite-ground station communication to be configured for encrypted transmission of telecommands and status telemetry (TC/TM) through a satellite network
- Communication of confidential mission data by a satellite network to be encrypted by encryption solutions developed or verified as secure by the communication parties



- Communication of non-confidential mission data by a satellite network to be encrypted by encryption algorithms or modules developed or verified as secure by the communication parties

### Verification of Communication Integrity

- Integrity of communication between satellites and ground stations (exchange of command signals) to be verified by MAC<sup>6)</sup> or other means of communication integrity verification
- Order identifiers, time stamps, or random numbers to be used to prevent replay attacks

### Authentication Between Space Systems

- As satellite-ground station communication permitting access by unauthorized devices or systems is susceptible to mission data leaks or failure, mutual authentication is to precede such communication
- A function granting communication access to certain satellites and ground stations

### Security Measures in the Use of Foreign Ground Stations

- In instances requiring the use of foreign ground station services for increased utility of a space system, satellite network security measures to be formed from development stages to prevent foreign ground stations from taking control of satellites or accessing mission data

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6) MAC : Message Authentication Code

- Connection to foreign ground stations to be established by a dedicated line or VPN, with end-to-end encrypted communication of control signals and mission data taking place

### 3. Cybersecurity Measures for Satellites

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#### Response to Abnormal Commands

- Response to invalid commands to be prepared for instances where an attacker bypasses or affects authentication to hack a satellite
- Development of a function which, when a set of control commands that can cause a satellite to malfunction is received, makes a log entry without executing the commands
- Development of a “blacklisting” function which blocks systems or users after receiving from them a certain number of abnormal commands

#### Reception of Multiple Satellite Navigation Systems

- As satellite malfunction can result from satellite navigation system jamming, location and time data to be received from multiple reliable satellite navigation systems, such as GPS and Galileo

#### Secure Decommissioning of Satellites

- Decommissioning of satellites to include the prevention of data leaks and inter-satellite collisions
  - Permanent shutdown of key functions, e.g. satellite communication functions

- Permanent deletion of critical data, e.g. mission data, settings, logs
- Deletion of encryption keys and algorithms

### Control of Unnecessary Services and Functions

- Programs and services not required for satellite operation to be removed by checking their influence on satellite security in development stages
- Minimum permissions for necessary programs and services

### Log Management

- Creation and management of logs for efficiency of satellite operations and the management of incidents
- Preservation of logs for a minimum of one year, or transfer to a ground station (satellite control center) or deletion on a regular basis if storage capacity is limited
  - Encryption (guarantee of integrity) of logs before transfers

### Encryption of Critical Data

- Verifiable encryption of control commands, mission data, and other critical data for storage and transmission

### Security Measures for Satellite Operation in Safety Mode

- In preparation for instances requiring a satellite to activate a safety mode to avert issues, such essential security functions to be maintained as to prevent an attacker from taking control of the satellite or changing its operating mode for establishing minimum administrative security measures

## Section 2

## Cybersecurity measures in the operation and utilization phase

### Administrative Security Measures (Applicable to All Space Systems)

#### 1. Designation and Management of Protected Areas

To achieve high levels of security in the operation of ground stations, locations requiring physical protection are to be designated as protected areas and controlled in physical access.

##### Control of Outside Information and Communications Devices

- The presence of outside information and communications devices (laptop computers, portable storage devices, etc.) in protected areas is to be controlled.
  - Installation of only programs needed for work as approved by the CISO
  - Advance scanning of information and communications devices brought into protected areas for malware and unnecessary programs
  - Logging of information and communications devices brought into and removed from protected areas
  - No connection of outside information and communications devices to a ground station system

- Installation and use of necessary outside programs only on specified laptop computers or portable storage devices following all due security measures, including CISO's approval and scanning for malware

### Authorization and Identification of Persons on Entry/Exit

- Operation of 24/7 security-monitored entries and exits capable of the authorization and identification of persons entering and exiting to limit the access of protected areas to persons involved in the operation and use of ground stations only
- Biometric devices (fingerprint recognition) are recommended for personnel authorization and identification, or ID cards can be used as an alternative
- Operation of a single main point of entry/exit (the one always used) equipped with personnel authorization/identification devices

### Prevention of Data Leaks

- Prevention of data leaks from protected areas by the planning of secure data export and emergency deletion, and the prevention of photography /filming

## 2. Security Management for Outsourcing companies

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The use of outsourcing companies in the operation and use of ground stations, satellite networks, or satellites is to be preceded by security clearance of the participating persons and equipment, and the formation of product security measures. Control measures are to be implemented around space systems accessible by outsourcing companies to prevent alterations and data theft.

## Control of Outsourcing companies' Work Areas

- In-house setup of outsourcing companies' work areas to observe all due security measures
- Separation of information and communications networks used in outsourcing companies' work areas from the space system operator's information and communications networks
  - In instances requiring the access of the space system operator's information systems by outsourcing companies, the grant of limited access to only the necessary information systems from only the designated terminals, combined with access logging
- No unauthorized information and communications devices allowed in outsourcing companies' work areas; authorized devices to be port-sealed and security-stickered

## Control of Outsourcing Companies' Remote Works

- No development or maintenance activities from remote locations allowed; in unavoidable instances, the CISO to issue approval and implement due security measures beforehand
  - Permission of connection from designated terminals only, with the designated terminals allowed connection to only the necessary space systems through the space system operator's outsourcing company online security control system
  - No use of designated terminals for unauthorized purposes; operating systems to be kept updated; no installation of unnecessary programs
  - Encryption of communication in remote work; activation of communication only during work hours, followed by immediate deactivation

- The space system operator to issue advance approval of the particulars and duration of remote work to take place
- Particulars and history of remote work to be monitored

## Equipment Security Management

- The space system operator to provide and log equipment to be used by outsourcing companies
- In instances requiring the use of a outsourcing company's own equipment, the space system operator to take all due security steps (inspection and reset of equipment) before approving and logging the use
  - Equipment with components banned from use (wireless LAN equipment) on site to be identified and quarantined
- Physical sealing of ports and installation of security systems to prevent the use of unauthorized storage devices and communications devices

## Security Training and Inspection of Outsourcing Companies

- Obtaining of security agreements, training of outsourcing companies in security protocols, and regular and unscheduled, frequent inspection of security compliance for the identification and correction of issues
  - No copying or release to the outside of the space system operator's data
  - No work-related equipment to be brought onto or removed from work sites without authorization
  - No connection to unauthorized networks from smartphones

### 3. Management of Space User Accounts

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Space system user accounts are to be created only for persons who require one for work, with all due security measures implemented.

- Creation of an account for approved persons only; only one account person (no account sharing)
- Created accounts to provide essential permissions only; permissions to be regularly reviewed and updated
  - Immediate deletion or change of accounts in keeping with changes in personnel (transfers, exits, change of duties, etc.)
- All accounts are to be set up with a password, a combination of numbers, letters, and symbols that is difficult to guess and regularly changed.

### 4. Identification and Management of Space System Assets

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In the stages of space system operation and use, any changes in hardware or software assets are to be identified and logged. Movable assets such as communication terminals are susceptible to theft, loss, and damage so should be regularly checked.



## 5. Coordinated Response to Space System Cybersecurity Crises

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Response to a cybersecurity crisis occurring in the operation and use of space systems is to take place through close cooperation among the appropriate entities and in accordance with the crisis response procedures in place in each country.

- Working-level response to a cybersecurity crisis is to take place in the early stages of the crisis occurrence and include the isolation of the attacked network, removal of security vulnerabilities, and deletion or protection of the appropriate data
  - Notification of the the relevant entities on internal findings
  - Recovery from failures through backup and other measures
- Coordinate response to take place: Update of the relevant entities on developments, emergency communication network activation, alerts on prevention and response measures

## 6. Cybersecurity Training and Crisis Response Training of Space System Personnel

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An internally followed manual on crisis response, including the organization for crisis response, emergency communication network, and recovery procedures, is to be created based on a given entity's characteristics and kept updated. Crisis response training in various scenarios of crises is to take place regularly in accordance with the manual.

## Technical Security Measures

Safe operation of space systems such as ground stations and satellites requires the application and management of technical security measures.

### 1. Cybersecurity Measures for Ground Stations

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#### Control of Administrator Permissions

- Ground station system administrator accounts to be created and operated separately
- Connection to administrator services permitted only from designated administrator terminals assigned specific IP addresses

#### Updates and Security Patches

- Regular update and security patch of operating systems and software comprising ground station systems
  - Ground stations physically separated from the networks required for updates and security patches to be updated or security-patched using optical disks or dedicated portable storage devices
  - Use of a dedicated portable storage device to be preceded using a clean PC to scan the portable storage device for malware before connection to a ground station system
- Security patches to be first tested on a test system to verify security before application to actual systems; this minimizes system errors and failures

- Alternatives to be utilized in instances where a security patch cannot be applied due to compatibility or availability issues
  - Regular check of logs to identify issues
  - Other security measures as deemed necessary

### Control of Unnecessary Services and Functions

- Influences on the security of ground stations to be identified and disabled or removed
  - Unused servers and client programs
  - Software compilers
  - Backup files and databases used only for system development
  - All unused data and configuration files
  - Sample programs and scripts
- Minimum permissions for necessary programs and services

### Detection and Removal of Malware

- Installation of vaccine programs on servers, PCs, and other IT infrastructure in ground stations; regular update of vaccine programs for a regular scanning for security hazards
- Transfer of outside data to space systems to be preceded by scanning for malware using a network integration solution or clean computer
- Special security measures in instances where vaccine programs cannot be installed or updated

- Regular check of logs to identify issues
- Other security measures as deemed necessary

### Blocking of Connection of Unauthorized Storage Devices

- Prevention of the use of unauthorized portable storage devices (USB drives) by the physical blocking of ports or the use of information security solutions (device control systems)
- No auto opening of portable storage devices on connection; portable storage devices to be scanned for malware before use
- Logs on portable storage device connections to be preserved for a minimum of one year

### Enhanced Email Security

- Computers and servers connected to ground systems to be protected from email potentially containing malware with a malicious email blocking solution
- Settings to be made to prevent auto execution of attachments to received emails; attachments to be scanned for hidden malware using a vaccine program before execution
- No connection to external networks through links included in emails
- Designation of email accounts for communication of important information concerning space systems; definition of allowed senders and receivers; encryption of emails and attachments

### Protecting Critical Data

- Encryption or Access Control for Critical Data (mission data) stored in ground stations to reduce the risk of data leaks in the event of a security breach

### Backup of Settings and Stored Data

- Regular backup of settings and stored data on equipment used at ground stations; secure management of backup data to prevent damage or loss

### Log Management

- Preservation and management of logs for a minimum of one year for the efficient control and management of space systems, and for the tracking of incidents
  - Connected entities: Users, information systems, application programs, etc.
  - Types and times of activity: Log-on/-off, viewing/printing of data, etc.
  - Activity results: Successful/failed connections, etc.

### Reception of Multiple Satellite Navigation Systems

- Location and time data to be received from multiple reliable satellite navigation systems, such as GPS and Galileo, and checked for soundness

## 2. Cybersecurity Measures for Satellite Networks

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For security reasons, a satellite network is to be divided into subnetworks according to function, with communication among individual subnetworks being controlled in access and made secure with network integration security solutions.

## Ground Station Network Configuration and Management

### ① Satellite Control Centers

- Satellite control centers require cybersecurity measures commensurate with their role in the control of satellites, e.g. operation of satellites, mission planning, orbit setting
  - Connection to be established by one-way communication in instances where connection to an internal or external network is required for satellite operation data access (Fig. 3-1)
  - Data transfers from control networks to internal or external networks only security measures to be in place to prevent change of data transfer direction

### ② Satellite Data Centers

- External connections allowing satellite data centers to collect data for satellite applications through satellite control centers to be controlled and monitored
- Collection of outside data from reliable sources only; use of network integration solutions for scanning for malware before internal transmission of data (Fig. 3-2)

### ③ Satellite Communication Centers

- As the purpose of satellite communication centers is receiving and forwarding communicated data (relaying of data), connection is to be permitted only for authenticated communication terminals
- Strict control of unauthorized connections between satellite networks and internal networks with the use of security solutions such as network integration solutions and firewalls
- Satellite networks to be separately installed in line with purpose, e.g. work, internet connection; in instances where a separate installation is not possible, adequate security measures to be implemented, e.g. encryption, VPN, etc.

## Ground Station Network Integration

- A ground station network requiring connection to a space system of another entity to be connected by a dedicated line, with connection by a VPN being an alternative
  - Transfer of data and files over established connections to take place using network integration solutions
  - Information security solutions (firewalls) to be applied to established connections to allow only permitted servers to be connected from the outside
  - Restriction of communication to information sharing and data transfers only
  - Sufficient user authentication, e.g. certificates, biometrics, terminal authentication, to take place according to the level of importance of data to be accessed
- Ground stations equipped with cybersecurity measures suitable for a control system (satellite control centers) to be connected by one-way communication (Fig. 3-1)
  - One-way network for sending and receiving data to be separately set up only in instances where data transmission to a satellite control network is required
  - A DMZ to be set up before connecting a satellite control network to an external network, with the connection being established by one-way outbound communication and one-way inbound communication
  - Systems requiring external connection (connection to an outside entity) for data transmission to be connected by two-way communication through a dedicated line or VPN
  - Systems requiring connection to the outside to be restricted to access of only servers and equipment on satellite control networks as necessary for the provision of data to the outside

## Encrypted Communication Configuration

- Satellite-ground station communication to be configured for encrypted transmission of telecommands and status telemetry (TC/TM)
- Communication of confidential mission data by a satellite network to be encrypted by encryption solutions developed or verified as secure by the communication parties
- Communication of non-confidential mission data by a satellite network to be encrypted by encryption algorithms or approved encryption modules developed or verified as secure by the communication parties

## Verification of Communication Integrity

- Integrity of communication between satellites and ground stations (exchange of command signals) to be verified by MACs or other means of communication integrity verification
- Order identifiers, time stamps, or random numbers to be used to prevent replay attacks

## Authentication Between Space Systems

- Mutual authentication is to precede satellite-ground station communication
- Satellites to accept communication signals (commands) from previously permitted or authenticated ground stations only
- Communication terminals to access authenticated satellite services only
- Detection and blocking of unauthorized access or abnormal behavior of authenticated devices



## Security Measures in the Use of Foreign Ground Stations

- Connection to foreign ground stations to be established by a dedicated line or VPN, with end-to-end encrypted communication of control signals and mission data taking place
  - Use of foreign ground stations to be limited to the communication of non-confidential mission data only; communication of confidential data to be preceded by a security briefing with the relevant entities
- Satellites to accept communication signals (commands) from previously permitted or authenticated foreign ground stations only; signals from unauthenticated foreign ground stations to be blocked

## 3. Cybersecurity Measures for Satellites

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### Response to Abnormal Commands

- Receipt of a set of control commands that can cause a satellite to malfunction to be followed by log entry without execution of the commands
- Implementation of a “blacklisting” function which blocks systems or users after receiving from them a certain number of abnormal commands

### Reception of Multiple Satellite Navigation Systems

- As satellite malfunction can result from satellite navigation system jamming, location and time data to be received from multiple reliable satellite navigation systems, such as GPS and Galileo

## Control of Unnecessary Services and Functions

- Programs and services not required for satellite operation to be disabled and removed by checking their influence on satellite security
- Minimum permissions for necessary programs and services

## Log Management

- Creation and management of logs for efficiency of satellite operations and the management of incidents
- Logs to be transferred to a ground station or deleted on a regular basis if storage capacity is limited
  - Encryption (guarantee of integrity) of logs before transfers

## Encryption of Critical Data

- Control commands, mission data, and other critical data to be encrypted before storage or transmission

## Security Measures for Satellite Operation in Safety Mode

- In preparation for instances requiring a satellite to activate a safety mode to avert issues, such essential security functions to be maintained as to prevent an attacker from taking control of the satellite or changing its operating mode through authentication or encryption

## Section 3

# Security Measures in Stages of Decommissioning

## Administrative Security Measures

### 1. Decommissioning of Space System-related Devices and Data

The appropriate security measures are to be implemented in the decommissioning of space system-related devices and data to prevent the exposure of mission data or operational data in storage.

- Deletion of data stored in storage devices to prevent exposure to the outside
- Complete destruction of storage devices holding secrets, restricted information, or encryption keys by incineration, pulverization, or dissolution
- In instances where the incineration, pulverization, or dissolution of a storage device in a space system located in space, e.g. satellites, is impossible, technical security measures to be implemented as suggested in this document.

## 2. Security Management of Mission-completed Satellites

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Continuing operation of a mission-completed space system (satellites) for purposes such as the maintenance of an orbit requires security management to prevent leaks of critical data or physical loss such as that resulting from inter-satellite collisions.

- Regular checks for the survival of the space system
- Regular inspections of logs for the identification and management of malicious activities

### Technical Security Measures

The decommissioning of mission-completed space systems (satellites) requires secure procedures to prevent leaks of critical data or physical loss such as that resulting from inter-satellite collisions.

Accordingly, a mission-completed space system can be physical destroyed by automated or manual means to prevent its involvement in malicious activities, or its battery can be fully depleted to prevent its malfunction.

The transfer of a space system to another party is to be preceded by the necessary security procedures to prevent leaks of critical data.

## 1. Secure Decommissioning of Space Systems

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### Physical Decommissioning

- Permanent shutdown of key functions, e.g. satellite communication functions
- Utilization of strategies such as incineration by reentry into Earth's atmosphere, or shift in an orbit for decommissioning

### Suspension of Functions

- Permanent deletion of critical mission data, settings, and logs
- Deletion of encryption keys and algorithms

### Authentication of Commands for the Suspension of Functions

- Commands for the suspension of functions to be received from designated ground stations and administrators only and authenticated to prevent the receipt of such commands from an attacker

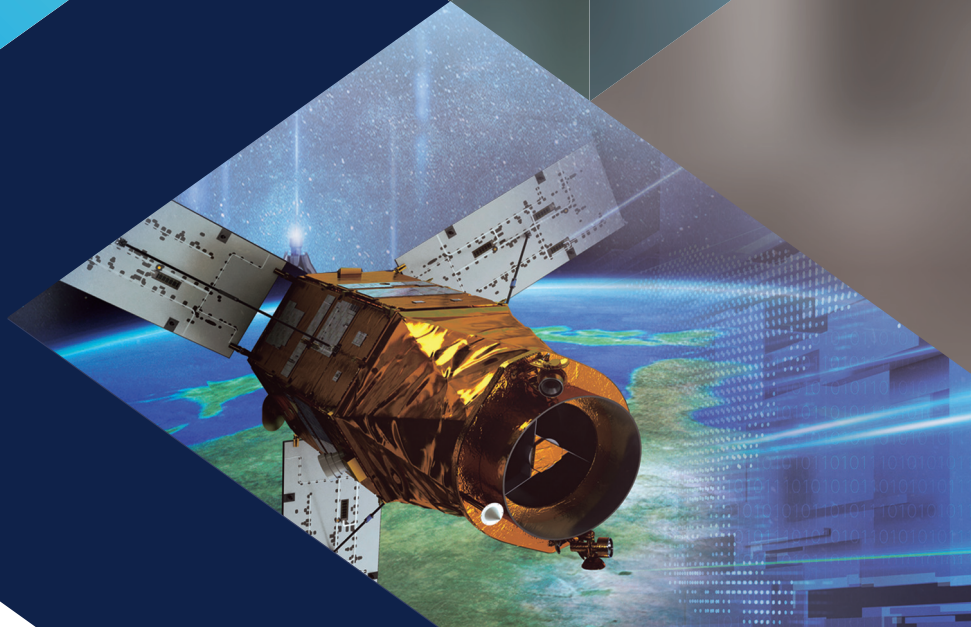
## 2. Transfer of Space Stations to Another Party

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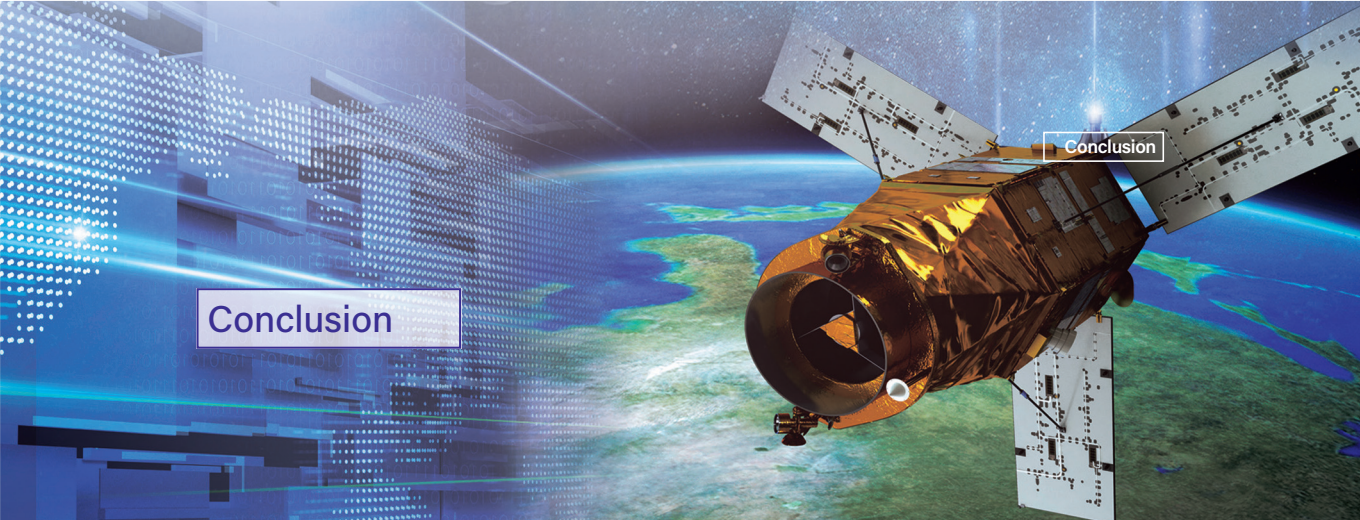
### Transfer of Ownership or Operating Rights

- Permanent deletion of critical mission data, settings, logs, and encryption keys and algorithms
- The relevant entities to be consulted in advance on the necessary security measures

## Conclusion







Low Earth orbit satellites and other such advanced space systems today make our hyper-connected society possible and keep it advancing still, and we must understand the importance of cybersecurity in improving our quality of life and enhancing national competitiveness. Only with such an understanding and attention can cybersecurity be improved.

The development of space systems necessitates a substantial investment, and the exchange of critical data heightens the potential consequences of any failures. Such failures could have a profound impact not only on economic stability but also on national security. For these reasons, it can be said that we have reached a point where responding to the growing cybersecurity threats to space systems has become crucial for organizations and individuals alike to take decisive action.

This document has suggested the essential security measures to be considered in the design and development, operation and use, and decommissioning of space systems. The task is to maintain the security of space systems such as ground stations, satellites networks, satellites, launch vehicles, and launchers throughout their life cycles. It is in the interests of a nation to establish a space system cybersecurity system for that purpose.

Without the due steps taken to establish and continuously improve the cybersecurity of national space systems, the hazards endured could include communication failures and threats to national security. The guidelines found in this document should be referenced in endeavors to improve cybersecurity to keep space systems steadfast in their purpose.





Appendix

# Cybersecurity Checklists for Space Systems







## Cybersecurity Checklists for Space Systems

| [Appendix 1] Ground Stations |                                                                   |                     |                        |                 |
|------------------------------|-------------------------------------------------------------------|---------------------|------------------------|-----------------|
| No                           | Item                                                              | Design /Development | Operation /Utilization | Decommissioning |
| 1                            | Planning of the cybersecurity for ground stations                 | ●                   |                        |                 |
| 2                            | Designation and management of protected areas                     | ●                   | ●                      |                 |
| 3                            | Security management for outsourcing companies                     | ●                   | ●                      |                 |
| 4                            | Management of ground system user accounts                         | ●                   | ●                      |                 |
| 5                            | Restriction on external disclosure of critical information        | ●                   |                        |                 |
| 6                            | Supply chain security management (hardware and software included) | ●                   |                        |                 |
| 7                            | Identification and management of ground station assets            | ●                   | ●                      |                 |
| 8                            | Cyber crisis response for ground systems                          | ●                   | ●                      |                 |
| 9                            | Cybersecurity training of ground station personnel                | ●                   | ●                      |                 |
| 10                           | Access control of ground station administrator permissions        | ●                   | ●                      |                 |
| 11                           | Updates and security patches for ground stations                  | ●                   | ●                      |                 |

| No | Item                                                                      | Design<br>/Development | Operation<br>/Utilization | Decommissioning |
|----|---------------------------------------------------------------------------|------------------------|---------------------------|-----------------|
| 12 | Restriction of unnecessary ground station services and functions          | ●                      | ●                         |                 |
| 13 | Detection and removal of malware at ground stations                       | ●                      | ●                         |                 |
| 14 | Blocking of connection of unauthorized storage devices at ground stations | ●                      | ●                         |                 |
| 15 | Enhanced email security of ground stations                                | ●                      | ●                         |                 |
| 16 | Encryption of critical data at ground stations                            | ●                      | ●                         |                 |
| 17 | Backup of settings and stored data at ground stations                     | ●                      | ●                         |                 |
| 18 | Management of ground station logs                                         | ●                      | ●                         |                 |
| 19 | Reception of multiple satellite navigation systems by ground stations     | ●                      | ●                         |                 |
| 20 | Decommissioning of ground station devices and data                        |                        |                           | ●               |
| 21 | Secure decommissioning measures for ground stations                       | ●                      |                           | ●               |
| 22 | Transfer of ground stations to another party                              |                        |                           | ●               |

| [Appendix 2] Satellite Networks |                                                                   |                     |                        |                 |
|---------------------------------|-------------------------------------------------------------------|---------------------|------------------------|-----------------|
| No                              | Item                                                              | Design /Development | Operation /Utilization | Decommissioning |
| 1                               | Planning of the cybersecurity for satellite networks              | ●                   |                        |                 |
| 2                               | Designation and management of protected areas                     | ●                   | ●                      |                 |
| 3                               | Security management for outsourcing companies                     | ●                   | ●                      |                 |
| 4                               | Management of satellite network user accounts                     | ●                   | ●                      |                 |
| 5                               | Restriction on external disclosure of critical information        | ●                   |                        |                 |
| 6                               | Supply chain security management (hardware and software included) | ●                   | ●                      |                 |
| 7                               | Identification and management of satellite network assets         | ●                   | ●                      |                 |
| 8                               | Cyber crisis response for satellite network                       | ●                   | ●                      |                 |
| 9                               | Cybersecurity training of satellite network personnel             | ●                   | ●                      |                 |
| 10                              | Access Control of satellite networks                              | ●                   | ●                      |                 |
| 11                              | Interconnection between satellite networks                        | ●                   | ●                      |                 |
| 12                              | Configuration of encrypted communication for satellite networks   | ●                   | ●                      |                 |
| 13                              | Verification of communication integrity for satellite networks    | ●                   | ●                      |                 |

| No | Item                                                   | Design<br>/Development | Operation<br>/Utilization | Decommissioning |
|----|--------------------------------------------------------|------------------------|---------------------------|-----------------|
| 14 | Authentication between satellite networks              | ●                      | ●                         |                 |
| 15 | Security measures in the use of global ground stations | ●                      | ●                         |                 |
| 16 | Decommissioning of satellite network devices and data  |                        |                           | ●               |
| 17 | Transfer of satellite networks to another party        |                        |                           | ●               |

| [Appendix 3] Satellites |                                                                   |                     |                        |                 |
|-------------------------|-------------------------------------------------------------------|---------------------|------------------------|-----------------|
| No                      | Item                                                              | Design /Development | Operation /Utilization | Decommissioning |
| 1                       | Planning of the cybersecurity for satellites                      | ●                   |                        |                 |
| 2                       | Designation and management of protected areas                     | ●                   | ●                      |                 |
| 3                       | Security management for outsourcing companies                     | ●                   | ●                      |                 |
| 4                       | Management of access accounts                                     | ●                   | ●                      |                 |
| 5                       | Restriction on external disclosure of critical information        | ●                   | ●                      |                 |
| 6                       | Supply chain security management (hardware and software included) | ●                   | ●                      |                 |
| 7                       | Identification and management of assets                           | ●                   | ●                      |                 |
| 8                       | Cyber crisis reponse for satellites                               | ●                   | ●                      |                 |
| 9                       | Cybersecurity training for personnel                              | ●                   | ●                      |                 |
| 10                      | Response to abnormal commands                                     | ●                   | ●                      |                 |
| 11                      | Reception of multiple GNSS signals                                | ●                   | ●                      |                 |
| 12                      | Secure decommissioning                                            | ●                   |                        | ●               |
| 13                      | Restriction of unnecessary services and functions                 | ●                   | ●                      |                 |
| 14                      | Management of logs                                                | ●                   | ●                      |                 |
| 15                      | Encryption of critical data                                       | ●                   | ●                      |                 |
| 16                      | Security measures for operation in safe mode                      | ●                   | ●                      |                 |



| No | Item                                             | Design<br>/Development | Operation<br>/Utilization | Decommissioning |
|----|--------------------------------------------------|------------------------|---------------------------|-----------------|
| 17 | Decommissioning and disposal of devices and data |                        |                           | ●               |
| 18 | Security management of end-of-life satellites    |                        |                           | ●               |
| 19 | Secure decommissioning measures for satellites   | ●                      |                           | ●               |
| 20 | Transfer of satellites to another party          |                        |                           | ●               |